Online Appendix

When Does Crime Enter House Prices? Evidence from Chicago

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A Data and Measurement

Table OA1: Sample Construction

	Transactions	Parcels
MLS sale listings (Chicago, SF/CO/RI)	1,243,680	
sold (closed transaction)	562,770	
closed 2007–2025	546,494	
price \$10,000–\$10,000,000	543,535	341,896
constant parcel count (analysis sample)	530,395	337,461
of which repeat-sale parcels (≥ 2)	329,800	136,866
identified by parcel + tract×month FE	242,815	

Notes: Chicago Multiple Listing Service closed transactions in single-family, condominium, and residential-income properties. Within-property identification comes from the 136,866 parcels that transact at least twice. Adding tract×month fixed effects to the parcel fixed effects drops transactions that are the only sale in their tract×month cell (and the singletons this cascade creates), leaving the 242,815 transactions that identify the two-way fixed-effect model. Crime-specific specifications are smaller still (the near–far regressor also requires within-parcel variation in exposure), and every sale-price regression uses the 242,815-sale repeat-sale estimating sample unless a table states otherwise.

Table OA2: Listings With vs Without Nearby Crime

	3-month window		6-month window	
	No crime ($N = 97,348$)	Any crime ($N = 433,017$)	No crime ($N = 47,022$)	Any crime ($N = 483,343$)
Close price (\$)	381,096	363,512	370,137	366,409
Log close price	12.54	12.42	12.53	12.43
Days on market	86	98	83	97
Any list-price cut	0.68	0.67	0.69	0.67
Bedrooms	3.1	3.0	3.0	3.0
Bathrooms	2.3	2.1	2.3	2.1
Living area (sq. ft.)	1,282	1,149	1,302	1,161
Total rooms	7.3	7.0	7.2	7.1
Year built	1954	1954	1955	1954

Notes: Sale analysis sample ($N = 530,395$; the 530,365 with an exposure classification are split) by whether any of the 27 offense groups records an incident within 0.05 miles in the 3 or 6 months before sale. Listings near recent crime are cheaper, smaller, and slower to sell; the raw price gap (Table 1) is confounded by these and unobserved differences, motivating the within-parcel, tract×month design.

Table OA3: Chicago Police Codes in Each Offense Group

Offense group	CPD primary type(s)	What is included (IUCR code)	Codes moved elsewhere	Incidents
Priced				
Homicide	Homicide	First degree murder (0110); reckless homicide (0142); involuntary manslaughter (0141); second degree murder (0130)	—	10,766
Narcotics	Narcotics; Other narcotic violation	Poss: cannabis 30gms or less (1811); poss: heroin(white) (2024); poss: crack (2027); manu/deliver: heroin (white) (2014); found suspect narcotics (2093); poss: cocaine (2022); and 43 other codes	—	439,526
Burglary	Burglary	Forcible entry (0610); unlawful entry (0620); attempt forcible entry (0630); home invasion (0650); burglary from motor vehicle (0760)	—	296,724
Vandalism	Criminal damage	To vehicle (1320); to property (1310); criminal defacement (1305); to city of chicago property (1345); to state sup prop (1340); institutional vandalism (1375); and 2 other codes	—	632,166
Suggestive				
Disorderly conduct	Public peace violation	Reckless conduct (0470); bomb threat (2850); arson threat (2851); other violation (2890); false police report (2860); mob action (3100); and 8 other codes	—	39,056
Sex offenses	Sex offense; Crim sexual assault; Criminal sexual assault; Offense involving children; Public indecency	Crim sexual assault: non-aggravated (0281); sex offense: criminal sexual abuse (1563); sex offense: public indecency (1570); offense involving children: agg crim sex abuse fam member (1752); crim sexual assault: aggravated: other (0265); sex offense: agg criminal sexual abuse (1562); and 27 other codes	Family (non-violent): 8 codes, e.g. child abuse (1750). Kidnapping: child abduction (1790). Child pornography/obscenity: 4 codes, e.g. child pornography (1582). Sex offenses (nonforcible): 4 codes, e.g. sex relation in family (1580)	63,549
Precise nulls				
Larceny/theft	Theft; Deceptive practice	Theft: $500 and under (0820); theft: over $500 (0810); theft: from building (0890); theft: retail theft (0860); deceptive practice: financial identity theft over $ 300 (1153); theft: pocket-picking (0870); and 22 other codes	Fraud: 21 codes, e.g. credit card fraud (1150). Forgery: 3 codes, e.g. forgery (1120). Bad checks: bogus check (1110). Stolen property: stolen prop: buy/receive/pos. (1200). And 1 other groups	1,364,912
Motor vehicle theft	Motor vehicle theft	Automobile (0910); attempt - automobile (0920); theft/recovery: automobile (0930); truck, bus, motor home (0915); cycle, scooter, bike w-vin (0917); theft/recovery: truck,bus,mhome (0935); and 6 other codes	—	292,290

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Table OA3, continued

Offense group	CPD primary type(s)	What is included (IUCR code)	Codes moved elsewhere	Incidents
Assault (non-dom.)	Battery; Assault	Battery: simple (0460); assault: simple (0560); assault: aggravated: handgun (051A); battery: aggravated: other dang weapon (0430); battery: aggravated: handgun (041A); battery: aggravated:knife/cutting instr (0420); and 32 other codes	—	739,961
Bad checks	Deceptive practice	Bogus check (1110)	Fraud: 21 codes, e.g. credit card fraud (1150). Larceny/theft: 13 codes, e.g. financial identity theft over $ 300 (1153). Forgery: 3 codes, e.g. forgery (1120). Stolen property: stolen prop: buy/receive/pos. (1200). And 1 other groups	9,424
Family (non-violent)	Offense involving children	Child abuse (1750); other offense (1780); endangering life/health child (1710); child abandonment (1755); contribute delinquency of a child (1720); harbor runaway (1791); and 2 other codes	Sex offenses: 4 codes, e.g. agg crim sex abuse fam member (1752). Kidnapping: child abduction (1790). Child pornography/obscenity: 4 codes, e.g. child pornography (1582)	29,131
Fraud	Deceptive practice	Credit card fraud (1150); fraud or confidence game (1130); illegal use cash card (1152); computer fraud (1242); finan exploit-elderly/disabled (1195); deceptive collection practices (1185); and 15 other codes	Larceny/theft: 13 codes, e.g. financial identity theft over $ 300 (1153). Forgery: 3 codes, e.g. forgery (1120). Bad checks: bogus check (1110). Stolen property: stolen prop: buy/receive/pos. (1200). And 1 other groups	136,957
Assault (domestic)	Battery; Assault	Battery: domestic battery simple (0486); assault: simple (0560); battery: aggravated domestic battery: other dang weapon (0497); assault: aggravated: handgun (051A); battery: simple (0460); battery: aggravated domestic battery: knife/cutting inst (0496); and 37 other codes	—	670,887
Trespass	Criminal trespass	To land (1330); to residence (1365); to state sup land (1350); to vehicle (1360); to airport (1335)	—	138,862
Robbery	Robbery	Armed: handgun (031A); strongarm - no weapon (0320); aggravated (0330); aggravated vehicular hijacking (0326); armed: other dangerous weapon (0313); armed:knife/cutting instrument (0312); and 8 other codes	—	213,364

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Table OA3, continued

Offense group	CPD primary type(s)	What is included (IUCR code)	Codes moved elsewhere	Incidents
Forgery	Deceptive practice	Forgery (1120); counterfeiting document (1121); counterfeit check (1122)	Fraud: 21 codes, e.g. credit card fraud (1150). Larceny/theft: 13 codes, e.g. financial identity theft over $ 300 (1153). Bad checks: bogus check (1110). Stolen property: stolen prop: buy/receive/pos. (1200). And 1 other groups	32,711
Liquor law	Liquor law violation	Liquor license violation (2250); illegal possession by minor (2220); sell/give/del liquor to minor (2210); illegal consumption by minor (2230); minor misrepresent age (2240); employ minor (2251)	—	7,941
Prostitution	Prostitution	Solicit on public way (1506); solicit for business (1513); solicit for prostitute (1512); call operation (1505); other prostitution offense (1549); patronizing a prostitute (1525); and 8 other codes	—	31,173
Inconclusive				
Weapons	Weapons violation; Other offense; Concealed carry license violation	Weapons violation: unlawful poss of handgun (143A); weapons violation: reckless firearm discharge (1477); other offense: other weapons violation (5007); weapons violation: unlawful use handgun (141A); weapons violation: unlawful use other dang weapon (141C); other offense: gun offender: annual registration (5111); and 19 other codes	Other offenses: 35 codes, e.g. telephone threat (2820). Arson: 4 codes, e.g. other arson/explosive incident (5003). Stolen property: internet offenses - electronic fencing (1147); internet offenses - intent of online sale of stolen property (1145)	114,818
Arson	Arson; Other offense	By fire (1020); attempt arson (1090); aggravated (1025); other arson/explosive incident (5003); by explosive (1010); pos: explosive/incendiary dev (1030); and 4 other codes	Other offenses: 35 codes, e.g. telephone threat (2820). Weapons: 7 codes, e.g. other weapons violation (5007). Stolen property: internet offenses - electronic fencing (1147); internet offenses - intent of online sale of stolen property (1145)	9,488
Gambling	Gambling	Game/dice (1661); other (1680); game/amusement device (1670); game/cards (1651); illegal ill lottery (1626); bookmaking/sports (1611); and 10 other codes	—	8,105

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Table OA3, continued

Offense group	CPD primary type(s)	What is included (IUCR code)	Codes moved elsewhere	Incidents
Stolen property	Deceptive practice; Other offense	Stolen prop: buy/receive/pos. (1200); internet offenses - electronic fencing (1147); internet offenses - intent of online sale of stolen property (1145)	Other offenses: 35 codes, e.g. telephone threat (2820). Fraud: 21 codes, e.g. credit card fraud (1150). Larceny/theft: 13 codes, e.g. financial identity theft over $ 300 (1153). Forgery: 3 codes, e.g. forgery (1120). And 4 other groups	1,586

Notes. Each offense group is a set of Illinois Uniform Crime Reporting (IUCR) codes used by the Chicago Police Department; offense groups follow Table A3. "What is included" lists the IUCR descriptions in the group with their codes, all of them when there are at most 6 and otherwise the 6 most frequent in 2007–2025. "Codes moved elsewhere" lists codes of the same police primary type that are assigned to a different group. Incidents are all reported incidents in 2007–2025 and equal those in Table 1. (a) Domestic and non-domestic assault are not separated by IUCR code: assault and battery incidents are split by the department's domestic flag on each incident report. (b) Two sparse groups are not estimated: human trafficking (140 incidents) and nonforcible sex offenses (82). (c) Family (non-violent) is the department's "offense involving children" primary type without its pornography, abduction, and sexual codes; it includes child-endangerment and child-abuse codes, so "non-violent" is a label, not a claim about every incident. (d) Source: City of Chicago Data Portal, *Crimes – 2001 to Present*.

Table OA4: White-Collar Offenses and the Financial-Magnitude Account

Offense	Identifying parcels	Pooled 2007–25	Crisis 2009–13	Recovery 2014–25
Fraud	67,030	−0.11	−0.10	−0.10
		(0.08)	(0.18)	(0.08)
Forgery	23,614	0.21	−0.04	0.16
		(0.17)	(0.66)	(0.17)
Bad checks	8,919	0.66**	1.60**	0.42
		(0.33)	(0.75)	(0.47)

Notes. White-collar offenses, estimated on the same specification as Table 2 ($\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$; near 0–0.05 against 0.20–0.30 miles; parcel and tract × year-month fixed effects, clustered by tract; N = 242,815). The financial-magnitude account predicts that offenses inflicting large monetary losses capitalize into prices. Embezzlement has too few incidents to estimate. Each offense is shown at its main window as in Table 2 (twelve months for offenses without a significant discount); identifying parcels are counted at that window; standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Additional Sale-Price Estimates

Estimates as fixed effects are added. Table OA5 reports the near–far estimate as the specification adds structure. Most of the raw association between near-ring crime and price reflects where and when properties sell: tract-by-month effects alone remove most of the raw homicide

estimate. Observed characteristics are not a substitute for the parcel fixed effect. A cross-sectional hedonic regression overstates the discounts for burglary, narcotics, and vandalism, consistent with exposed properties differing in ways the listed characteristics do not record, and understates the homicide discount. Parcel fixed effects alone are not sufficient either. They absorb time-invariant differences across properties but not changes in the surrounding neighborhood between sales, and without tract-by-month effects the homicide estimate changes sign.

Table OA5: The Near–Far Discount Across Fixed-Effect Specifications

Offense	(0) Raw	(1) Tract×mo.	(2) + Char.	(3) Parcel FE	(4) Parcel FE + tract×mo.
Homicide	-12.81^{***}	-2.29	-2.63^{**}	0.24	-3.02^{**}
	(3.71)	(1.43)	(1.07)	(1.63)	(1.38)
Burglary	-3.50^{***}	-0.71^{***}	-0.32^{**}	-0.90^{***}	-0.24^{**}
	(0.50)	(0.27)	(0.15)	(0.21)	(0.12)
Narcotics	-2.14^{***}	-0.42^{***}	-0.51^{***}	-1.00^{***}	-0.29^{**}
	(0.54)	(0.16)	(0.13)	(0.31)	(0.14)
Vandalism	-3.52^{***}	-1.18^{***}	-0.48^{***}	-0.31^{**}	-0.17^{**}
	(0.59)	(0.22)	(0.13)	(0.15)	(0.08)
Tract×month FE	No	Yes	Yes	No	Yes
Characteristics	No	No	Yes	No	No
Parcel FE	No	No	No	Yes	Yes

Notes. Each cell is a separate near–far regression of ln(sale price) on near (0–0.05 mi) and far (0.20–0.30 mi) incident counts; $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ is the percent price change per near-ring incident, standard error below. Columns: (0) no fixed effects; (1) tract×year-month effects; (2) adds observed listing characteristics (bedrooms, bathrooms, ln living and lot area, age, units, stories, property type, new construction); (3) parcel fixed effects in place of characteristics; (4) parcel and tract×year-month fixed effects, the baseline specification. All five columns use the *same* rows, the column-(4) estimating sample (N = 242,815 repeat-sale transactions); columns (1)–(4) also include the stale-crime persistence terms. Each offense is at its main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months). Clustered by tract.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA6: Near-Only Specification and the Far-Ring Coefficient, All Twelve Windows

	Exposure window (cumulative months)											
Specification	1	2	3	4	5	6	7	8	9	10	11	12
Homicide												
Near only (no far-ring terms)	−4.31	−1.27	−0.70	−1.44	−0.63	−0.71	−1.08	−0.83	−2.24*	**−2.83****	−2.84**	−2.17*
	(3.15)	(2.41)	(2.31)	(1.91)	(1.77)	(1.74)	(1.50)	(1.39)	(1.32)	(1.35)	(1.33)	(1.30)
Baseline δ	−5.86*	−2.23	−1.45	−2.04	−1.28	−1.30	−1.59	−1.24	−2.54*	**−3.02****	−3.15**	−2.49*
	(3.35)	(2.60)	(2.46)	(2.05)	(1.82)	(1.77)	(1.53)	(1.43)	(1.36)	(1.38)	(1.38)	(1.36)
Baseline β_{far}	1.92**	1.22*	0.98*	0.79	0.84*	0.80*	0.66	0.52	0.37	**0.20**	0.40	0.41
	(0.88)	(0.73)	(0.59)	(0.53)	(0.50)	(0.46)	(0.43)	(0.39)	(0.36)	(0.35)	(0.34)	(0.33)
Narcotics												
Near only (no far-ring terms)	−0.94	−0.49	−0.43	−0.40	−0.36	−0.25	−0.26	−0.41**	−0.34**	**−0.32****	−0.36***	−0.35***
	(0.71)	(0.41)	(0.37)	(0.27)	(0.23)	(0.19)	(0.17)	(0.17)	(0.15)	(0.14)	(0.13)	(0.10)
Baseline δ	−0.93	−0.37	−0.35	−0.32	−0.28	−0.18	−0.20	−0.38**	−0.31**	**−0.29****	−0.35***	−0.33***
	(0.75)	(0.42)	(0.38)	(0.27)	(0.23)	(0.19)	(0.17)	(0.18)	(0.15)	(0.14)	(0.13)	(0.11)
Baseline β_{far}	−0.01	−0.15*	−0.11	−0.10*	−0.10**	−0.08*	−0.08**	−0.03	−0.03	**−0.03**	−0.01	−0.02
	(0.13)	(0.08)	(0.07)	(0.05)	(0.05)	(0.04)	(0.04)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Burglary												
Near only (no far-ring terms)	−0.06	−0.19	−0.18	−0.23	−0.27*	−0.36***	−0.33***	**−0.27****	−0.25**	−0.25***	−0.25***	−0.21**
	(0.30)	(0.20)	(0.17)	(0.16)	(0.14)	(0.14)	(0.12)	(0.11)	(0.11)	(0.10)	(0.09)	(0.09)
Baseline δ	−0.20	−0.23	−0.20	−0.21	−0.25*	−0.34**	−0.30**	**−0.24****	−0.22**	−0.22**	−0.22**	−0.18*
	(0.31)	(0.20)	(0.17)	(0.16)	(0.15)	(0.14)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.09)
Baseline β_{far}	0.14*	0.05	0.02	−0.02	−0.02	−0.02	−0.04	**−0.04**	−0.04	−0.04	−0.04	−0.03
	(0.08)	(0.06)	(0.05)	(0.05)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)
Vandalism												
Near only (no far-ring terms)	0.31	0.14	−0.03	−0.09	−0.15*	**−0.16***	−0.16**	−0.13*	−0.10	−0.08	−0.06	−0.06
	(0.21)	(0.14)	(0.12)	(0.10)	(0.09)	(0.08)	(0.08)	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)
Baseline δ	0.26	0.13	−0.06	−0.11	−0.16*	**−0.17****	−0.16**	−0.13*	−0.10	−0.08	−0.06	−0.05
	(0.22)	(0.15)	(0.12)	(0.10)	(0.09)	(0.08)	(0.08)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)
Baseline β_{far}	0.05	0.01	0.03	0.02	0.01	**0.01**	−0.00	−0.00	−0.00	−0.01	−0.01	−0.01
	(0.06)	(0.04)	(0.04)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)

Notes. Pre-sale exposure; estimates in percent, standard errors clustered by census tract in parentheses; parcel and tract×month fixed effects; $N = 242{,}815$. *Near only* regresses log sale price on the near-ring count and its persistence control, with no far-ring terms. *Baseline* is the specification of Table 2: $\delta = \beta_{\text{near}} - \beta_{\text{far}}$ and the far-ring coefficient β_{far} (0.20–0.30 miles). Bold: each offense's main window. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Number of nearby incidents. Table OA7 and Figure OF1 report the count-indicator estimates. For homicide at ten months, the single-incident estimate is negative but imprecisely estimated (−2.22 percent, s.e. 1.43). The four-or-more discount is significant in 8 of 12 windows for burglary, 3 of 12 for vandalism, one for narcotics, and none for homicide, whose four-or-more category contains only 14 sales at ten months, compared with 6,577 for narcotics.

Table OA7: Dose-Response for All Estimable Offenses

Offense	1	2	3	4+	1	2	3	4+
	δ: near incidents (vs. 0)				listings identifying			
Homicide	−2.22	−20.17*	8.86	9.24	2,749	145	15	14
	(1.43)	(10.72)	(27.74)	(6.06)				
Burglary	0.48*	−0.51	−0.68	−1.85**	55,036	20,475	7,903	6,598
	(0.25)	(0.36)	(0.59)	(0.76)				
Narcotics	−0.34	0.36	−2.06*	−0.74	28,007	7,872	3,206	6,577
	(0.36)	(0.69)	(1.12)	(1.24)				
Vandalism	0.21	−0.45	−0.39	−1.05**	62,935	28,364	13,049	12,067
	(0.22)	(0.30)	(0.45)	(0.52)				

Notes. Price effect (%) of 1, 2, 3, and 4+ near (0–0.05 mi) incidents relative to zero, for all 22 estimable offenses. Parcel and tract × month fixed effects, pooled 2007–2025, clustered by tract; each offense at its main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months; 12 months for other offenses). The final four columns ("listings") give the number of sale listings identifying each count (not parcels); "–" denotes no listings at that count.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA8: When the Discount Appears: Non-Cumulative Exposure

Offense	Earliest sig. month	Late block (8–11 mo)	Concentrated in
Homicide	m1: −5.86*	−5.95** (2.91)	late (8–11 mo)
Burglary	m6: −0.85***	−0.09 (0.15)	early
Narcotics	m8: −2.09***	−0.64** (0.27)	late (8–11 mo)
Vandalism	m3: −0.44**	0.12 (0.10)	early

Notes. Non-cumulative decomposition of the near-ring discount by the lag at which crime occurs. "Earliest sig. month" is the first single month (crime in that month alone, $\delta = \beta_{\text{near},w} - \beta_{\text{near},w-1}$) at which the discount is significant; "Late block" is crime in the disjoint 8–11 month span entered as a single regressor (standard error in parentheses). Same specification as Table 2: parcel and tract × year-month fixed effects, clustered by tract, $N = 242{,}815$. Estimated jointly, the early and late blocks are statistically distinguishable only for vandalism (Figure A1); see Section 5.1.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

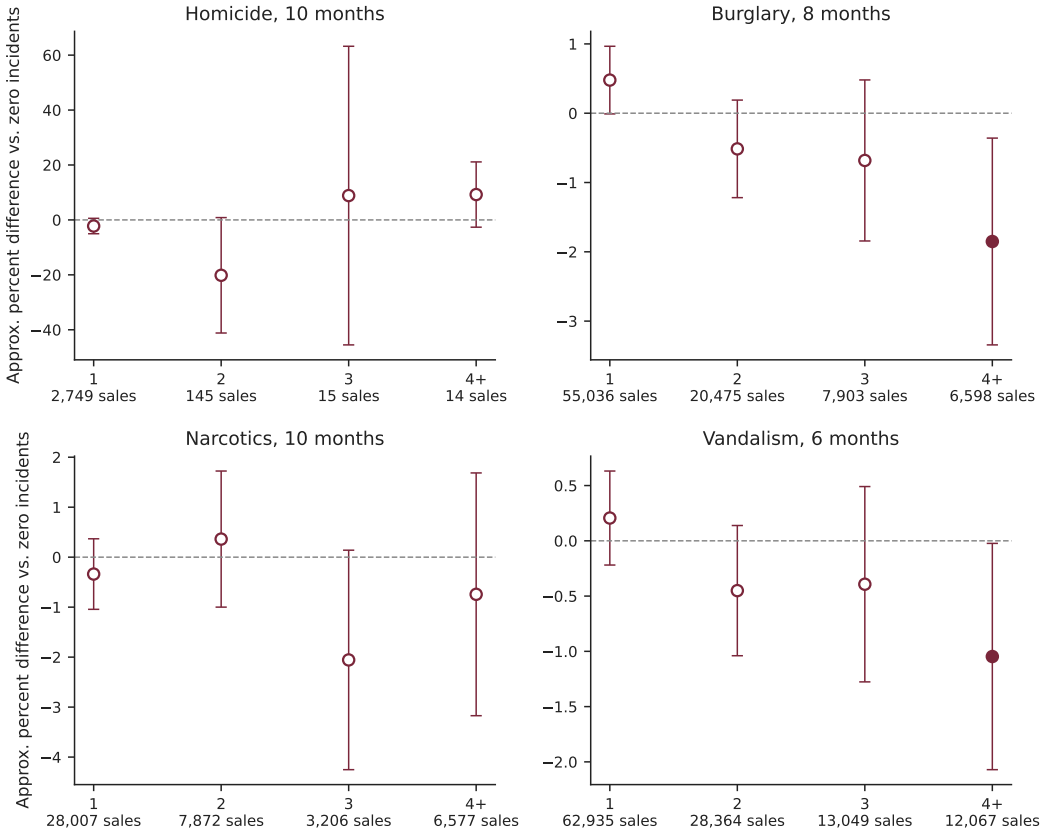


Figure OF1: Sale-price differences by number of nearby incidents. Each point is the estimated approximate percent difference in sale price ($100\hat{\beta}$ from the log-price regression) for one, two, three, or four or more incidents within 0.05 miles, relative to zero incidents, at each offense's main window (homicide and narcotics 10 months, burglary 8, vandalism 6). Estimates are total category effects, not effects per incident. Parcel and tract × year-month fixed effects; far-ring and persistence controls; standard errors clustered by tract. Counts below each category are sales in the estimation sample. Bars show 95% confidence intervals; filled markers indicate significance at the 5% level, hollow markers otherwise. Vertical scales differ across panels.

Table OA9: Sale-Price Capitalization by Domestic Status, All Offenses

Window (months)	Dom δ_{D}	Non-dom δ_{N}	Wald $H_0\colon \delta_{\mathrm{D}} = \delta_{\mathrm{N}}$
1	0.20	0.05	0.634
	(0.33)	(0.03)	
2	0.05	0.03	0.933
	(0.23)	(0.02)	
3	0.00	0.03	0.885
	(0.18)	(0.02)	
4	0.03	0.01	0.904
	(0.15)	(0.01)	
5	0.06	0.00	0.646
	(0.13)	(0.01)	
6	0.01	−0.01	0.910
	(0.11)	(0.01)	
7	−0.02	0.00	0.882
	(0.10)	(0.01)	
8	−0.06	0.00	0.547
	(0.10)	(0.01)	
9	−0.14	0.00	0.143
	(0.09)	(0.01)	
10	−0.13	0.00	0.145
	(0.09)	(0.01)	
11	−0.17*	0.00	0.062*
	(0.09)	(0.01)	
12	−0.17**	−0.01	0.046**
	(0.08)	(0.01)	

Notes. Near–far price effect $\delta = (\beta_{\mathrm{near}} - \beta_{\mathrm{far}}) \times 100$, the percent sale-price change per near-ring incident (crime within 0–0.05 miles against 0.20–0.30 miles), at each cumulative exposure window from one to twelve months. Column δ_{D} is the domestic-related arm (per the Illinois Domestic Violence Act) and δ_{N} the non-domestic arm, both included in the same regression; the last column is the p-value of the Wald test that the two are equal. Incidents are pooled across the twenty-two named offense groups and split by the CPD domestic flag (1,003,504 domestic vs 4,357,006 non-domestic geocoded incidents, 2007–2025). Baseline specification: parcel and tract × year-month fixed effects, clustered by census tract; $N = 242{,}815$ repeat-sale transactions, pooled 2007–2025. Standard errors in parentheses. Incident counts come from the geocoded 2007–2025 records with the domestic flag and differ by about 1.4 percent from the incident counts in Table 1, which use the exposure-construction pool.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA10: Sale-Price Capitalization of Domestic versus Non-Domestic Assault

Window (months)	Dom δ_{D}	Non-dom δ_{N}	Wald $H_0\colon \delta_{\mathrm{D}} = \delta_{\mathrm{N}}$
1	0.03	−0.03	0.896
	(0.37)	(0.20)	
2	0.01	−0.01	0.964
	(0.26)	(0.14)	
3	−0.02	0.03	0.851
	(0.21)	(0.11)	
4	0.02	−0.02	0.834
	(0.17)	(0.09)	
5	0.03	−0.03	0.748
	(0.15)	(0.08)	
6	0.02	−0.03	0.756
	(0.13)	(0.07)	
7	−0.02	−0.03	0.946
	(0.12)	(0.07)	
8	−0.08	−0.02	0.638
	(0.12)	(0.06)	
9	−0.16	−0.01	0.242
	(0.11)	(0.05)	
10	−0.13	0.01	0.261
	(0.10)	(0.05)	
11	−0.17	−0.02	0.195
	(0.10)	(0.04)	
12	−0.18*	−0.03	0.174
	(0.10)	(0.04)	

Notes. Near–far price effect $\delta = (\beta_{\mathrm{near}} - \beta_{\mathrm{far}}) \times 100$, the percent sale-price change per near-ring incident (crime within 0–0.05 miles against 0.20–0.30 miles), at each cumulative exposure window from one to twelve months. Domestic and non-domestic assault (split by the Chicago Police Department's domestic flag, based on the Illinois Domestic Violence Act) are separate near–far exposures in the same regression: column δ_{D} is the domestic assault arm, δ_{N} the non-domestic assault arm, and the last column the p-value of the Wald test that the two are equal. Parcel and tract × year-month fixed effects, clustered by census tract; N = 242,815 repeat-sale transactions, pooled 2007–2025. Standard errors in parentheses.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA11: Listing-Agent Local Experience and the Asking Price, All Windows

	Exposure window (cumulative months)											
Offense	1	2	3	4	5	6	7	8	9	10	11	12
Priced												
Homicide	−0.73	−0.08	−0.03	−0.32	−0.79*	−0.84**	−0.82**	−0.83**	−0.89**	−0.97***	−0.88***	−0.81***
	(0.89)	(0.65)	(0.52)	(0.45)	(0.43)	(0.41)	(0.38)	(0.36)	(0.35)	(0.34)	(0.32)	(0.31)
Burglary	0.03	0.02	0.02	0.05	0.06	0.01	0.03	0.00	−0.01	−0.01	−0.02	−0.00
	(0.09)	(0.07)	(0.06)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)
Narcotics	−0.09	−0.05	−0.08	−0.09*	−0.10*	−0.11**	−0.10***	−0.07**	−0.07**	−0.08***	−0.07***	−0.07***
	(0.12)	(0.08)	(0.07)	(0.05)	(0.05)	(0.05)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)
Vandalism	−0.02	−0.04	−0.03	−0.03	−0.04	−0.03	−0.02	−0.03	−0.04	−0.05**	−0.06**	−0.06**
	(0.07)	(0.05)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Suggestive												
Disorderly conduct	0.12	−0.08	−0.04	−0.22	−0.13	−0.12	−0.07	−0.05	−0.05	−0.08	−0.07	−0.07
	(0.24)	(0.20)	(0.17)	(0.16)	(0.13)	(0.12)	(0.11)	(0.10)	(0.10)	(0.09)	(0.09)	(0.09)
Sex offenses	0.12	−0.01	0.03	0.07	−0.02	−0.00	0.01	−0.03	−0.05	−0.06	−0.08	−0.08
	(0.18)	(0.14)	(0.11)	(0.09)	(0.09)	(0.08)	(0.08)	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)
Precise nulls												
Theft	−0.01	−0.01	−0.01	−0.01*	−0.00	−0.00	−0.00	−0.00	−0.00	−0.00	−0.00	−0.01
	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Motor vehicle theft	0.17**	0.03	0.02	0.01	0.04	0.02	0.04	0.03	0.02	0.02	0.03	0.02
	(0.08)	(0.06)	(0.05)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)
Robbery	−0.07	−0.14	−0.18**	−0.17**	−0.17***	−0.19***	−0.22***	−0.20***	−0.19***	−0.19***	−0.17***	−0.16***
	(0.11)	(0.09)	(0.08)	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Assault (non-domestic)	−0.07	−0.11***	−0.10***	−0.05*	−0.04	−0.05**	−0.06***	−0.07***	−0.07***	−0.08***	−0.07***	−0.07***
	(0.05)	(0.04)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.01)
Assault (domestic)	−0.25***	−0.15***	−0.15***	−0.11***	−0.07*	−0.04	−0.05	−0.05	−0.07**	−0.08***	−0.06**	−0.06**
	(0.08)	(0.06)	(0.05)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Criminal trespass	0.02	−0.01	−0.03	0.02	0.03	0.01	−0.01	−0.00	0.02	0.01	0.02	0.01
	(0.11)	(0.07)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Fraud	−0.07	−0.08*	−0.05	−0.09**	−0.08**	−0.07**	−0.07***	−0.07***	−0.05**	−0.05**	−0.04**	−0.04**
	(0.07)	(0.05)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Prostitution	−0.21	−0.03	0.08	0.08	0.10	0.07	−0.02	−0.03	−0.03	−0.02	0.00	−0.03
	(0.24)	(0.18)	(0.14)	(0.10)	(0.08)	(0.07)	(0.06)	(0.05)	(0.04)	(0.03)	(0.04)	(0.03)
Forgery	−0.06	−0.19	−0.20*	−0.16	−0.13	−0.14	−0.14	−0.12	−0.13	−0.15*	−0.16*	−0.14
	(0.19)	(0.13)	(0.12)	(0.11)	(0.10)	(0.10)	(0.10)	(0.10)	(0.09)	(0.09)	(0.08)	(0.09)
Family offenses	−0.15	0.00	−0.15	−0.20	−0.19	−0.18	−0.24	−0.19	−0.15	−0.17	−0.09	−0.07
	(0.44)	(0.29)	(0.24)	(0.22)	(0.19)	(0.18)	(0.17)	(0.16)	(0.16)	(0.16)	(0.15)	(0.15)
Bad checks	−0.50	−0.16	−0.08	−0.15	−0.14	−0.23	−0.29**	−0.26**	−0.22*	−0.17*	−0.14	−0.15
	(0.31)	(0.23)	(0.20)	(0.18)	(0.17)	(0.15)	(0.13)	(0.13)	(0.12)	(0.10)	(0.10)	(0.10)
Liquor law	−0.65*	−0.45*	−0.18	−0.23	−0.20	−0.06	−0.08	−0.07	−0.10	−0.12	−0.15	−0.13
	(0.37)	(0.26)	(0.21)	(0.19)	(0.17)	(0.15)	(0.15)	(0.14)	(0.13)	(0.13)	(0.12)	(0.11)
Inconclusive												
Weapons	0.14	0.13	0.21**	0.15	0.17*	0.12	0.11	0.11	0.09	0.13**	0.12**	0.11*
	(0.21)	(0.14)	(0.10)	(0.10)	(0.09)	(0.09)	(0.08)	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)
Arson	0.65	0.12	0.12	0.13	−0.07	−0.03	0.13	−0.04	0.01	0.09	0.10	0.05
	(0.74)	(0.60)	(0.52)	(0.42)	(0.40)	(0.38)	(0.34)	(0.33)	(0.32)	(0.31)	(0.30)	(0.29)
Gambling	−1.61	−1.46	−1.14	−0.26	−0.41	−0.71	−0.85	−0.90	−1.05*	−1.09**	−1.05**	−1.00**
	(1.58)	(1.02)	(0.82)	(0.80)	(0.69)	(0.69)	(0.65)	(0.63)	(0.57)	(0.54)	(0.50)	(0.49)
Stolen property	−0.26	−0.18	−0.68	−0.95	−0.82	−0.65	−0.66	−0.67	−0.42	−0.51	−0.51	−0.33
	(1.08)	(0.89)	(0.87)	(0.66)	(0.63)	(0.54)	(0.50)	(0.48)	(0.46)	(0.42)	(0.41)	(0.39)

Notes. Local-experience gradient, (near × E − far × E) × 100, at every cumulative window, where $E = \ln(1 +$ prior same-agent same-tract listings) is the listing agent's local experience, under the specification of Table 5: parcel, tract × month, and agent fixed effects; standard errors clustered by tract in parentheses; N = 972,629. Offense groups as in Table A3. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA12: Buyer's-Agent Local Experience and the Closing Price, All Windows

	Exposure window (cumulative months)											
Offense	1	2	3	4	5	6	7	8	9	10	11	12
Priced												
Homicide	−6.27**	−2.46	−2.04*	−1.29	−0.95	−1.06	−1.44*	−1.58**	−1.47**	−1.61**	−1.84**	−2.19***
	(2.60)	(1.73)	(1.18)	(1.21)	(0.88)	(0.81)	(0.78)	(0.78)	(0.70)	(0.75)	(0.79)	(0.81)
Burglary	0.53**	0.22	0.05	0.02	0.12	0.10	0.08	0.09	0.10	0.15	0.13	0.12
	(0.27)	(0.19)	(0.16)	(0.15)	(0.13)	(0.13)	(0.11)	(0.10)	(0.11)	(0.09)	(0.09)	(0.08)
Narcotics	−0.06	0.37	0.38*	0.32*	0.34**	0.18	0.11	0.10	0.09	0.08	0.06	0.10
	(0.47)	(0.29)	(0.22)	(0.19)	(0.16)	(0.14)	(0.13)	(0.11)	(0.09)	(0.10)	(0.10)	(0.08)
Vandalism	−0.04	0.09	0.06	0.06	0.11	0.14**	0.13**	0.09*	0.10*	0.09*	0.04	0.03
	(0.14)	(0.12)	(0.09)	(0.08)	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Suggestive												
Disorderly conduct	−0.15	−0.11	0.06	−0.18	−0.33	−0.42	−0.44**	−0.24	−0.34*	−0.48**	−0.42**	−0.47**
	(0.57)	(0.47)	(0.35)	(0.29)	(0.30)	(0.26)	(0.22)	(0.21)	(0.19)	(0.19)	(0.21)	(0.21)
Sex offenses	−0.49	−0.20	0.08	0.18	0.30	0.19	0.15	0.16	0.15	0.09	0.03	0.03
	(0.52)	(0.36)	(0.28)	(0.25)	(0.26)	(0.23)	(0.21)	(0.18)	(0.16)	(0.15)	(0.15)	(0.14)
Precise nulls												
Theft	0.02	−0.01	−0.01	−0.01	−0.00	−0.00	0.01	0.01	0.01	0.00	0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Motor vehicle theft	0.39*	0.48***	0.32**	0.32***	0.24**	0.10	0.05	0.07	0.03	0.07	0.05	0.00
	(0.22)	(0.14)	(0.13)	(0.12)	(0.10)	(0.09)	(0.09)	(0.07)	(0.07)	(0.07)	(0.07)	(0.06)
Robbery	0.43	0.29	0.20	0.16	0.14	0.08	−0.00	0.05	−0.04	0.02	0.03	0.03
	(0.28)	(0.20)	(0.17)	(0.16)	(0.11)	(0.12)	(0.10)	(0.10)	(0.09)	(0.08)	(0.07)	(0.07)
Assault (non-domestic)	0.45***	0.14	0.14*	0.12*	0.13*	0.16**	0.10*	0.11**	0.06	0.04	0.04	0.01
	(0.15)	(0.09)	(0.08)	(0.07)	(0.07)	(0.06)	(0.05)	(0.05)	(0.04)	(0.04)	(0.03)	(0.03)
Assault (domestic)	0.15	−0.12	−0.18	−0.14	−0.13	−0.13	−0.05	0.01	−0.01	0.02	−0.01	−0.05
	(0.27)	(0.17)	(0.13)	(0.11)	(0.08)	(0.08)	(0.07)	(0.07)	(0.06)	(0.07)	(0.05)	(0.05)
Criminal trespass	0.17	0.25	0.18	0.24**	0.16	0.03	0.05	0.02	−0.00	−0.00	0.03	0.05
	(0.25)	(0.16)	(0.12)	(0.10)	(0.11)	(0.11)	(0.10)	(0.11)	(0.10)	(0.09)	(0.07)	(0.06)
Fraud	−0.09	−0.08	−0.07	−0.12	−0.08	−0.11	−0.08	0.00	0.01	0.00	0.05	0.01
	(0.32)	(0.13)	(0.09)	(0.08)	(0.09)	(0.10)	(0.08)	(0.07)	(0.05)	(0.05)	(0.04)	(0.04)
Prostitution	1.37***	0.26	−0.19	−0.24	−0.01	−0.09	−0.08	−0.13	−0.09	−0.07	−0.10	−0.13
	(0.43)	(0.19)	(0.19)	(0.15)	(0.14)	(0.22)	(0.17)	(0.15)	(0.11)	(0.10)	(0.10)	(0.08)
Forgery	−0.38	−0.15	0.14	0.17	0.14	0.05	−0.08	−0.10	−0.18	−0.19	−0.23*	−0.21*
	(0.58)	(0.38)	(0.29)	(0.26)	(0.20)	(0.18)	(0.15)	(0.13)	(0.13)	(0.13)	(0.12)	(0.12)
Family offenses	−0.59	0.15	−0.25	−0.18	−0.01	0.20	0.12	−0.01	0.13	0.15	0.06	0.11
	(1.16)	(0.85)	(0.79)	(0.62)	(0.53)	(0.41)	(0.37)	(0.39)	(0.37)	(0.36)	(0.32)	(0.30)
Bad checks	−0.60	0.33	−0.17	−0.36	−0.43	−0.10	−0.01	−0.00	−0.16	−0.14	−0.25	−0.25
	(1.67)	(0.91)	(0.65)	(0.62)	(0.63)	(0.56)	(0.50)	(0.41)	(0.38)	(0.37)	(0.30)	(0.27)
Liquor law	1.39	1.34	1.21	0.76	0.19	0.01	−0.33	0.06	0.00	−0.07	−0.11	0.28
	(1.89)	(1.13)	(1.04)	(0.85)	(0.76)	(0.74)	(0.56)	(0.60)	(0.55)	(0.51)	(0.46)	(0.42)
Inconclusive												
Weapons	0.18	0.00	−0.23	−0.31	−0.17	−0.08	−0.14	−0.07	0.03	0.02	0.08	0.11
	(0.50)	(0.30)	(0.27)	(0.29)	(0.18)	(0.15)	(0.13)	(0.12)	(0.12)	(0.10)	(0.09)	(0.08)
Arson	4.46**	2.06	2.08	2.74*	2.82**	2.31**	2.07*	1.82*	2.15**	1.65*	1.87**	1.93**
	(1.90)	(1.57)	(1.43)	(1.44)	(1.25)	(1.17)	(1.12)	(1.05)	(0.99)	(0.88)	(0.83)	(0.84)
Gambling	14.04	6.27	3.18	3.75	4.63	3.61	3.06	2.84	1.78	0.50	−0.56	−0.82
	(16.26)	(5.96)	(3.73)	(3.70)	(3.33)	(2.99)	(2.54)	(2.24)	(2.08)	(1.85)	(1.81)	(1.80)
Stolen property	3.22	4.50	0.81	0.23	−0.62	−2.29	−1.63	−1.31	−1.53	−1.20	−0.56	0.12
	(7.08)	(2.78)	(2.86)	(3.07)	(2.52)	(2.23)	(2.18)	(1.97)	(1.75)	(1.48)	(1.27)	(1.16)

Notes. Local-experience gradient, (near × E − far × E) × 100, at every cumulative window, where $E = \ln(1 +$ prior same-agent same-tract listings) is the buyer's agent's local experience, under the specification of Table 5: parcel, tract × month, and agent fixed effects; standard errors clustered by tract in parentheses; N = 186,510. Offense groups as in Table A3. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Incident Location Classification

Several tests split an offense's incidents by *where* they occur. I classify each incident using the Chicago Police Department's `location_description` field into three mutually exclusive categories. *Commercial* covers places of business: banks, currency exchanges, and ATMs; retail, grocery, and drug stores; restaurants, bars, and hotels; gas stations; commercial, medical, and dental offices; factories and warehouses; and entertainment venues. *Residential* covers the dwelling interior and its curtilage: residences, apartments, and houses, together with garages, porches, yards, hallways, stairwells, basements, and public-housing units. *Street* covers the public way and other outdoor, encounterable space: streets, sidewalks, alleys, and gangways; parking and vacant lots; highways, parks, and waterfronts; and vehicles and transit. Incidents whose location maps to none of these, blank or ambiguous entries, are dropped, roughly one in eight. The street category is the encounterable bucket, because it is the space a passer-by can observe during a visit to the block.

This classification underlies the incident-location split (Table A7).

Applying the same split to the four priced offenses (Table A7) localizes the discount differently across them. A component significant on its own is not necessarily distinguishable from the others, so I report the Wald tests that the residential component equals the street and commercial components. The homicide and narcotics discounts are significant for incidents on the street (in 4 and 5 of 12 windows) but not for incidents at residential addresses; the two estimates differ significantly only for homicide at the twelve-month window ($p = 0.046$); the vandalism discount is concentrated at residential addresses (8 of 12 windows) and differs significantly from street vandalism in 7 of 12 ($p = 0.018$ at the main window), with a post-sale placebo indistinguishable from zero. Burglary's near–far discount does not localize cleanly in this coarse residential–street–commercial split: neither its pooled residential nor its street component is distinguishable from the other, and the pooled residential coefficient is matched window-for-window by post-sale crime. The finer within-residential split in Table 3 is sharper: burglary's visible-*frontage* component is large and significant (-0.74, s.e. 0.33, significant in 7 of 12 windows) with a clean placebo (-0.20, $p = 0.45$), while the hidden interior is null (-0.14, s.e. 0.16), so the pooled residential placebo failure traces to the interior subcategory, not to the frontage incidents that carry the effect. Several additional offenses show negative residential coefficients in isolation, but those patterns are mirrored by post-sale crime and disappear when the pre-sale effect is compared directly with its placebo; the priced set is therefore not an artifact of the binary $p < .05$ placebo screen. Table A9 reports the split for every estimable offense (the priced-four summary is Table A7).

Because the reported estimates fix each offense at a single main window, Table OA13 reports the cross-location Wald tests at every exposure window, so the location heterogeneity can be assessed across the full window range rather than a selected one. The residential-versus-street difference for vandalism is significant across seven consecutive windows; homicide crosses conventional significance only at the twelve-month window, where its residential estimate is wrongly signed; burglary's only significant contrasts are residential-versus-commercial at ten and eleven months, driven by a residential component that fails the post-sale placebo; and narcotics never rejects. The vandalism

result is therefore not a selected-window artifact, while the street concentration for homicide and narcotics is descriptive rather than a demonstrated cross-location difference.

Table OA13: Cross-Location Wald Tests by Exposure Window, Priced Offenses

Window	Homicide	Narcotics	Vandalism	Burglary
Panel A. Residential = street/public (p)				
1	0.227	0.447	0.495	0.216
2	0.099*	0.647	0.270	0.520
3	0.526	0.520	0.238	0.916
4	0.624	0.948	0.182	0.712
5	0.678	0.647	0.090*	0.925
6	0.410	0.843	0.018**	0.810
7	0.297	0.853	0.002***	0.425
8	0.136	0.679	0.001***	0.289
9	0.106	0.851	0.002***	0.340
10	0.137	0.897	0.001***	0.317
11	0.080*	0.879	0.002***	0.340
12	0.046**	0.677	0.000***	0.287
Panel B. Residential = commercial (p)				
1	0.380	0.148	0.605	0.847
2	0.147	0.689	0.383	0.273
3	0.498	0.411	0.417	0.545
4	0.689	0.865	0.551	0.227
5	0.475	0.625	0.475	0.294
6	0.420	0.826	0.258	0.158
7	0.500	0.960	0.233	0.362
8	0.345	0.845	0.252	0.251
9	0.817	0.975	0.234	0.121
10	0.916	0.871	0.113	0.034**
11	0.717	0.861	0.154	0.032**
12	0.379	0.550	0.142	0.104

Notes. Wald p-values that the residential near–far component equals the street/public component (Panel A) and the commercial component (Panel B), at every exposure window 1–12 months before the sale, for the four priced offenses. Pooled 2007–2025, parcel and tract×month fixed effects, standard errors clustered by tract.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA14: Residential Location Codes and Their Frontage/Interior Assignment

CPD location code	Incidents	Main	Strict
Residence	1,401,849	Interior	Interior
Apartment	1,032,507	Interior	Interior
Residence-Garage	134,912	Frontage	Frontage
Residence Porch/Hallway[†]	124,149	Frontage	Dropped
Residential Yard (Front/Back)	75,130	Frontage	Frontage
Cha Apartment	40,920	Interior	Interior
Driveway - Residential	25,198	Frontage	Frontage
Cha Hallway/Stairwell/Elevator	25,021	Frontage	Dropped
Residence - Porch / Hallway	19,974	Frontage	Dropped
Residence - Yard (Front / Back)	17,113	Frontage	Frontage
Residence - Garage	16,299	Frontage	Frontage
House	740	Interior	Interior
Porch	415	Frontage	Dropped
Yard	342	Frontage	Frontage
Hallway	111	Frontage	Dropped
Garage	75	Frontage	Frontage
Cha Hallway	40	Frontage	Dropped
Vestibule	28	Frontage	Dropped
Stairwell	28	Frontage	Dropped
Cha Stairwell	11	Frontage	Dropped
Cha Lobby	9	Frontage	Dropped
Trailer	4	Interior	Interior
Cha Breezeway	3	Frontage	Dropped
Coach House	3	Interior	Interior
Rooming House	2	Interior	Interior

Notes. Every residential-address value of the Chicago Police Department `location_description` field, its total incident count over 2001–2025, and its assignment in the within-residential frontage/interior split (Table 3). *Main* is the baseline definition: frontage (the visible outdoor curtilage and building-exterior/entry codes) versus the dwelling interior. *Strict* restricts frontage to outdoor curtilage only (garage, yard, driveway, gangway) and drops the fused porch/hallway code and interior common areas; estimates under the strict definition are not reported.

[†]The fused `PORCH/HALLWAY` code combines a visible porch with a hidden hallway; it is counted as frontage in the main definition and dropped in the strict definition.

The location split could also reflect objective risk: residential and street incidents may differ in how well they predict future crime. Table A10 regresses future near-ring crime on past surrounding crime separately by incident location, without fixed effects, as a buyer would observe it at the sale. For vandalism, the two forecast future near-ring crime about equally well at six months ($R^2 = 0.148$ for residential and 0.125 for street), yet only residential vandalism is priced. The pricing difference therefore does not follow from a difference in recurrence, although I do not formally test the small gap in R^2. For homicide and narcotics, street incidents forecast better, and for burglary residential incidents do, consistent with where these offenses occur.

D Target of the Incident

Table 3, Panel B holds observability roughly fixed (both targets are frontage incidents, visible from the street) and varies what the offense harms, splitting vandalism into damage to the property and damage to a vehicle. Table A11 reports the full twelve-window version under both fixed-effect designs. Under parcel and tract×year-month fixed effects the damage-to-property arm is negative and significant from the seventh exposure month (-0.89 to -1.03) while the damage-to-vehicle arm is null at every window, and the two differ most at the twelve-month window ($p = 0.057$). The two detailed police offense codes forecast their own recurrence at the same small rate (the Table A12 design applied to each; ρ between -0.00 and -0.02, indistinguishable, $z < 1$), so the contrast holds the recurrence benchmark fixed and varies only what is harmed. Under parcel×era fixed effects both arms are negative and of similar size and are not distinguishable at any window; this specification drops to about half the sample ($N = 123{,}133$), because parcel×era is identified only from parcels that resell within the same era, so it is lower-powered and its flat arms are consistent with, not evidence against, the parcel-and-month split.

Table OA15: Within-Property Visibility, All Estimable Offenses (Window-by-Window)

Offense	Frontage δ [min, max]	Fr. sig$^-$ (plac.)	Interior δ [min, max]	Int. sig$^-$	Fr. $\neq$ Int.
Burglary	[−0.76, −0.37]	7/12 (0)	[−0.30, −0.11]	0/12	1/12
Narcotics	[−7.45, −2.35]	7/12 (0)	[−1.20, 0.92]	0/12	6/12
Vandalism	[−0.60, 1.02]	3/12 (0)	[−0.79, −0.34]	9/12	3/12
Homicide	[−23.69, 2.04]	0/12 (2)	[−0.65, 6.31]	0/12	0/12
Larceny/theft	[−0.09, 0.17]	0/12 (0)	[−0.08, 0.45]	0/12	0/12
Motor-vehicle theft	[0.52, 2.33]	0/12 (0)	[1.02, 5.35]	0/12	0/12
Robbery	[−2.41, 1.90]	0/12 (0)	[0.57, 3.89]	0/12	5/12
Assault (non-dom.)	[−0.43, 0.34]	0/12 (0)	[−0.49, 0.02]	1/12	0/12
Assault (dom.)	[−1.14, 1.89]	0/12 (6)	[−0.24, 0.12]	0/12	0/12
Sex offenses	[−4.27, 1.98]	0/12 (0)	[−0.61, 0.12]	1/12	0/12
Prostitution	[−9.87, 10.85]	0/12 (6)	[0.62, 3.31]	0/12	0/12
Disorderly conduct	[−0.81, 3.83]	0/12 (0)	[−2.07, −0.98]	2/12	0/12
Trespass	[−0.79, 0.03]	1/12 (0)	[−0.81, −0.28]	0/12	0/12
Weapons	[−10.15, −1.48]	0/12 (0)	[−3.67, 2.33]	0/12	1/12
Arson	[−10.05, 0.95]	0/12 (0)	[−5.58, 4.33]	0/12	0/12
Fraud	[−4.24, −1.07]	0/12 (0)	[−0.41, −0.18]	5/12	0/12
Forgery	[−2.08, 10.26]	0/12 (1)	[−1.11, 0.35]	0/12	0/12
Bad checks	[−1.29, 15.59]	0/12 (5)	[0.27, 1.65]	0/12	0/12
Family (non-violent)	[−0.98, 6.86]	0/12 (0)	[0.75, 2.59]	0/12	0/12
Gambling	[−19.36, 13.53]	0/12 (0)	[−35.03, 30.37]	1/12	0/12
Stolen property	[−6.42, 16.84]	0/12 (0)	[−17.80, 1.04]	0/12	0/12
Liquor law	[−20.27, 10.80]	1/12 (8)	[−0.25, 7.95]	0/12	1/12
All offenses (pooled)	[−0.23, 0.01]	5/12 (0)	[−0.14, 0.07]	4/12	0/12

Notes. Window-by-window companion to Table 3: residential incidents of each offense split between the visible frontage (garage, porch, yard, driveway, porch/hallway and common areas; Table OA14) and the hidden dwelling interior, estimated jointly, pooled 2007–2025, parcel and tract×month fixed effects, clustered by tract. For each offense the frontage and interior columns give the range of $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ across the twelve exposure windows; "sig$^-$" counts windows with a significant negative coefficient at five percent, and the parenthetical is the number in which the post-sale placebo (crime after the sale) is significant. The last column counts windows in which frontage differs from interior at five percent (Wald). Frontage narcotics is a small subsample (incidents recorded at the porch or garage), hence its large per-incident magnitudes and standard errors. For larceny, the frontage and interior estimates do not differ significantly ($p = 0.686$).

Table OA16: Relevance: What the Offense Threatens and Whether It Is Capitalized

Offense	What is threatened	Capitalized
Anchored to the address (stays with the property and its immediate surroundings)		
Burglary	dwelling	Yes
Homicide[‡]	place (violent event)	Yes
Narcotics	place (environment)	Yes
Vandalism	structure/land	Yes
Disorderly conduct	place (environment)	Suggestive
Arson	structure/land	No
Gambling	place (environment)	No
Liquor	place (environment)	No
Prostitution	place (environment)	No
Trespass[†]	structure/land	No
Weapons	place (environment)	No
Harms a person, money, or a movable item		
Sex offenses	person	Suggestive
Assault (domestic)	person	No
Assault (non-dom.)	person	No
Bad checks	financial	No
Family	person	No
Forgery	financial	No
Fraud	financial	No
Larceny/theft	portable asset	No
Motor vehicle theft	portable asset	No
Robbery	person	No
Stolen property	portable asset	No

Notes. Offenses classified by whether the harm stays with the address (dwelling, structure or land, or the place itself) or harms a person, money, or a movable item. "Capitalized" follows the offense sort in Table A2: *Yes* for the priced offenses, *Suggestive* for the suggestive tier, *No* otherwise.
[‡]Homicide is person-directed; it is placed here because its discount is statistically significant only for incidents in public space near the home, not for incidents at the dwelling. Were it classified as travelling with the household, it would be the one priced offense in that group.
[†]Trespass is imprecisely estimated: no discount is detected, but the estimate is consistent with a discount as large as 0.2 to 0.3 percent per incident.

E Additional Price-Formation Tables

This appendix collects supporting tables for Section 5.3. Table OA17 decomposes the asking-price experience gradient into the slope among experienced agents and the first-time-in-tract differential.

Table OA17: Experience Gradient vs First-Timer Discontinuity

Offense	Fixed effects	γ_E (experienced slope)	γ_0 (first-timer)
Narcotics	No agent FE	−0.12 (0.07)	−0.02 (0.10)
Narcotics	Agent FE	−0.05 (0.05)	0.02 (0.07)
Vandalism	No agent FE	−0.04 (0.03)	0.18^{***} (0.07)
Vandalism	Agent FE	−0.01 (0.03)	0.14^{**} (0.06)
Homicide	No agent FE	0.09 (0.74)	1.75 (1.61)
Homicide	Agent FE	−0.21 (0.56)	1.58 (1.25)

Notes. Asking-price experience gradient with a first-time-in-tract dummy (40.8% of listings), at the 12-month window, per unit of E. γ_E is the slope among agents with prior experience; γ_0 is the first-timer differential in the near–far ask. For narcotics γ_0 is null, but γ_E attenuates and is not significant once the first-timer differential is allowed for. For vandalism γ_E collapses to zero and γ_0 is positive and significant: first-time agents ask *more* (a discontinuity, not experience accumulating). Homicide is too sparse to separate the two. Standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA18: The Quantity Margin: Listings and Sales

Offense	P(listed) (pp)	P(sold) (pp)	Difference (pp)
Positive difference (listing response exceeds completed-sale response)			
Narcotics	0.02^{***}	0.00	0.03^{***}
	(0.01)	(0.00)	(0.00)
Burglary	0.06^{***}	0.02^{***}	0.04^{***}
	(0.01)	(0.01)	(0.01)
Arson	0.13^{**}	−0.01	0.14^{***}
	(0.05)	(0.04)	(0.04)
Assault (domestic)	0.00	-0.02^{***}	0.02^{***}
	(0.01)	(0.00)	(0.01)
Homicide	0.10^{**}	−0.04	0.14^{***}
	(0.05)	(0.04)	(0.04)
Vandalism	0.02^{***}	0.01	0.02^{***}
	(0.01)	(0.01)	(0.01)
Assault (non-dom.)	0.01	−0.01	0.01^{***}
	(0.01)	(0.01)	(0.00)
Weapons	0.03^{*}	−0.01	0.04^{***}
	(0.02)	(0.01)	(0.01)
Gambling	0.15^{**}	−0.02	0.17^{***}
	(0.07)	(0.05)	(0.06)
Disorderly conduct	0.08^{***}	0.04	0.04^{**}
	(0.03)	(0.03)	(0.02)
No difference			
Trespass	0.01	0.00	0.01
	(0.02)	(0.01)	(0.01)
Liquor law	0.00	−0.07	0.06
	(0.07)	(0.05)	(0.06)
Motor-vehicle theft	−0.02	−0.01	−0.01
	(0.02)	(0.01)	(0.01)
Family (non-violent)	0.05	0.02	0.02
	(0.03)	(0.03)	(0.02)
Forgery	0.06	0.03	0.03
	(0.04)	(0.03)	(0.03)
Bad checks	−0.05	−0.01	−0.04
	(0.05)	(0.05)	(0.06)
Stolen property	0.08	0.15	−0.07
	(0.14)	(0.12)	(0.11)
Fraud	−0.02	−0.01	−0.01
	(0.01)	(0.01)	(0.01)
Robbery	0.01	0.01	0.01
	(0.01)	(0.01)	(0.01)
Larceny/theft	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Prostitution	−0.02	-0.02^{**}	0.00
	(0.02)	(0.01)	(0.02)
Sex offenses	−0.01	−0.01	0.00
	(0.02)	(0.02)	(0.02)

Notes. Near–far effect on the annual probability a parcel is *listed* and that it *sells*, in percentage points (s.e. in parentheses; base rates 6.0 and 3.8 percent). P(sold) is the annual turnover rate at the parcel level, the object measured at the tract level by Boggess et al. (2013). The *difference* is listed minus sold, so a positive difference means the listing response exceeds the completed-sale response; it is not a direct withdrawal estimate (withdrawal: Table 4). Each annual parcel-year regression includes parcel and tract × year fixed effects and no other controls; near and far counts are incidents in the same calendar year as the listing or sale; standard errors are clustered by tract. $N = 13,682,888$ parcel-years.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F Rental Market

Table OA19: Rental Agent Local Experience (Letting Agent)

Offense	Discount	Experience grad./SD
Asking rent		
Sex offenses	-0.21^{**}	-0.048
	(0.09)	(0.065)
Prostitution	-0.15	-0.012
	(0.11)	(0.072)
Liquor law	-0.27^{***}	-0.021
	(0.10)	(0.059)
Burglary	-0.11^{*}	0.047
	(0.06)	(0.035)
Narcotics	0.10	-0.035^{*}
	(0.08)	(0.019)
Days on market		
Sex offenses	1.07	-0.085
	(0.75)	(0.632)
Prostitution	-2.00^{***}	0.922
	(0.69)	(1.029)
Liquor law	1.06	-0.011
	(1.81)	(0.860)
Burglary	0.68	0.322
	(0.46)	(0.451)
Narcotics	1.01	-0.586^{***}
	(0.72)	(0.215)
Withdrawal		
Sex offenses	-0.11	-0.407^{**}
	(0.32)	(0.185)
Prostitution	-0.60^{*}	0.325
	(0.34)	(0.340)
Liquor law	0.04	0.515
	(0.43)	(0.382)
Burglary	-0.04	0.049
	(0.19)	(0.117)
Narcotics	0.10	0.009
	(0.18)	(0.113)

Notes. Near–far effect on lease listings interacted with the letting (listing) agent's local experience $E = \ln(1 + \text{prior same-agent same-tract lease listings})$, standardized. Outcomes are ln(asking rent), ln(days on market), and P(withdrawal). Parcel×era + tract×month + agent (email) fixed effects, clustered by tract; each offense at a fixed window (burglary 8, sex offenses 9, prostitution 11, liquor 12, narcotics 12 months). "Discount" is the near–far effect at mean experience; "Experience grad./SD" is the change per SD of E. Standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

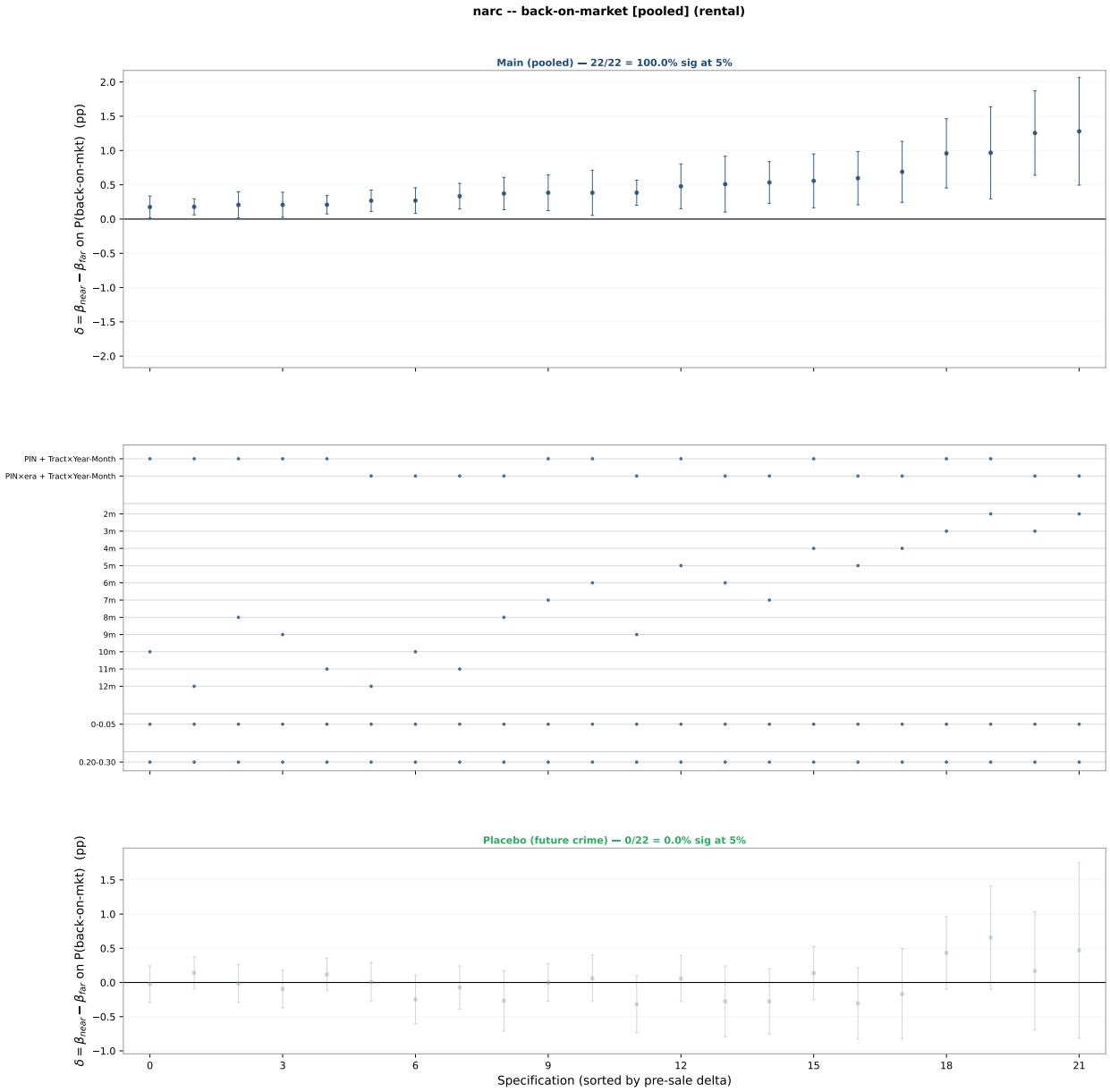


Figure OF2: Rental Leasing Friction: Relisting Response to Nearby Narcotics

Notes. Specification-curve "mirror" of the near–far effect ($\delta = \beta_{\text{near}} - \beta_{\text{far}}$, near 0–0.05 vs. far 0.20–0.30 mi) of nearby narcotics on the probability a lease listing is a relisting (the unit had an earlier spell on the market; percentage points), MLS lease listings, 2007–2025, clustered by tract. The top panel sorts the twenty-two specifications (eleven exposure windows, two to twelve months, × two fixed-effect designs: parcel + tract×month, and parcel×era + tract×month) by point estimate with 95% confidence intervals; the decoder panel identifies each specification; the bottom panel is the future-crime placebo (crime dated after the listing). Main: 22/22 significant at 5%; placebo: 0/22.

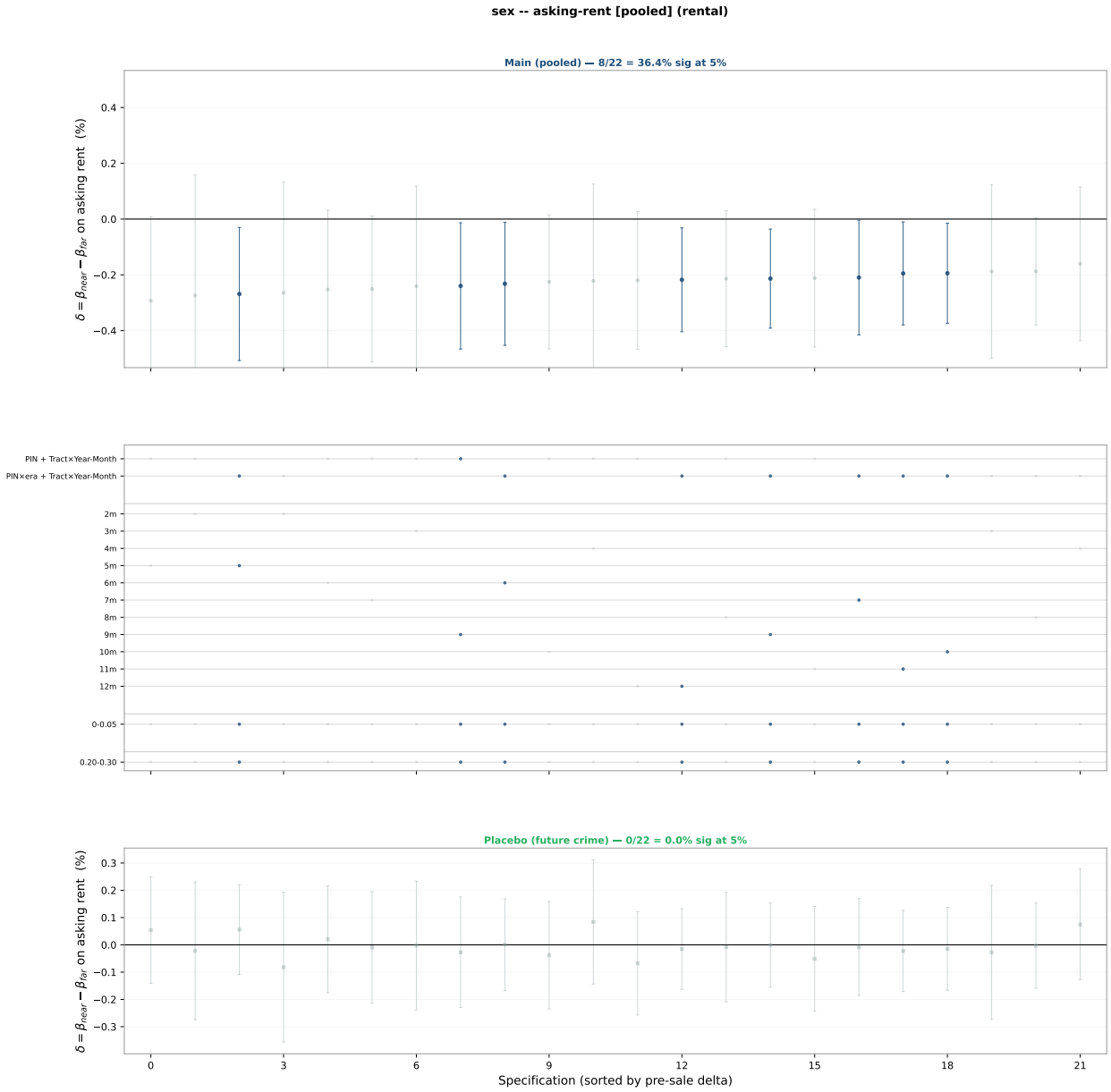


Figure OF3: Rental Asking-Rent Response to Nearby Sex Offenses

Notes. Specification-curve "mirror" of the near–far effect ($\delta = \beta_{\text{near}} - \beta_{\text{far}}$, near 0–0.05 vs. far 0.20–0.30 mi) of nearby sex offenses on ln(asking rent), percent, MLS lease listings, 2007–2025, clustered by tract. Twenty-two specifications (eleven exposure windows, two to twelve months, × parcel and parcel×era fixed effects), 95% confidence intervals; bottom panel is the future-crime placebo. Main: 8/22 significant at 5% (all in the discount direction); placebo: 0/22.

G Heterogeneity

The discounts also vary with neighborhood socioeconomic status. I use ten predetermined tract characteristics: median household income, the poverty rate, the share with a college degree, the owner-occupancy rate, median home value, median gross rent, the Black and Hispanic population shares, racial segregation, and the vacancy rate. In separate regressions, I interact each standard-

ized characteristic with the near–far term, controlling for the tract's predetermined crime level (Table OA20).

All four priced offenses carry larger discounts in lower-income and less-educated tracts; burglary and narcotics also carry larger discounts in higher-poverty tracts. A one-standard-deviation increase in tract income is associated with a 2.28-percentage-point smaller homicide discount, with smaller differences for burglary, narcotics, and vandalism. Discounts are likewise larger in tracts with lower median home values and rents, while neither owner-occupancy nor vacancy shows a consistent pattern. Income, poverty, education, home values, and rents are strongly correlated. The results suggest a common socioeconomic gradient rather than distinct effects of each characteristic. Table OA20 reports every offense and characteristic.

Discounts are also larger in tracts with a higher Black share. Controlling for predetermined tract crime, this gradient is marginally significant for homicide and burglary and significant for narcotics and vandalism. Hispanic share shows no significant gradient at the five-percent level for any priced offense (Table OA20). Because tracts with a higher Black share also tend to have lower incomes, I include income and racial composition jointly (Table OA21). With income and share Black together, both estimates lose precision for homicide, burglary, and narcotics, leaving their separate contributions unresolved. When income and racial segregation, measured by the divergence index (Roberto, 2016), are included jointly, the income gradient remains significant for burglary and narcotics, and at ten percent for homicide, while segregation shows no significant gradient. Vandalism is the exception: segregation remains significant (-0.56, s.e. 0.22), while income does not. This result may reflect a role for neighborhood stigma in the pricing of visible disorder, although the estimates do not establish that mechanism. Sampson and Raudenbush (2004) find that residents perceive more disorder in neighborhoods with more Black residents, holding observed disorder fixed.

Table OA20: SES Heterogeneity Gradients, All Offenses

Offense	Inc	Pov	Col	Own	HV	Rent	Blk	His	Seg	Vac
Homicide	2.28^{**}	-1.72	3.02^{**}	-0.43	2.11	1.82^{**}	-2.99^{*}	-0.11	-2.10	-0.56
Burglary	0.36^{***}	-0.43^{**}	0.52^{**}	0.02	0.40^{**}	0.30^{***}	-0.61^{*}	-0.40^{*}	-0.43	-0.13
Narcotics	0.26^{***}	-0.29^{***}	0.25^{***}	0.19	0.44^{***}	0.18^{**}	-0.27^{**}	0.06	-0.15	0.07
Vandalism	0.15^{**}	-0.04	0.31^{**}	-0.17	0.18	0.13^{**}	-0.46^{**}	-0.04	-0.60^{***}	-0.12
Theft	0.02^{**}	-0.04	0.04	-0.02	0.03	0.01^{*}	-0.02	-0.01	0.01	0.02
Motor-vehicle theft	-0.13	0.30^{**}	-0.12	-0.16	-0.10	-0.10	0.17	-0.05	0.01	0.00
Robbery	0.12	-0.34	0.24	0.46^{**}	0.03	0.14	0.05	-0.57^{**}	0.05	-0.08
Assault (non-dom.)	0.08^{*}	-0.11	0.21^{**}	0.00	0.12	0.08^{**}	-0.26^{*}	-0.10	-0.26^{*}	-0.03
Assault (dom.)	0.14	0.00	0.26^{**}	0.03	0.11	0.12^{*}	-0.15	-0.16	-0.09	-0.03
Disorderly conduct	0.29	-0.13	0.48	-0.49	0.23	0.29	-1.04	0.79	-0.81	0.18
Trespass	0.31^{***}	-0.64^{***}	1.02^{***}	0.15	0.58^{***}	0.19^{***}	-0.95^{***}	-0.74^{***}	-1.06^{**}	0.04
Prostitution	0.13	-0.09	0.22	0.07	0.08	0.02	-0.25	0.29	-0.77	0.15
Fraud	0.25^{**}	-0.37^{**}	0.63^{***}	0.20^{*}	0.23^{**}	0.19^{**}	-1.16^{***}	-0.40	-0.29	-0.07
Sex offenses	-0.09	-0.13	0.38	-0.17	-0.14	-0.12	-0.58	-0.33	-0.87	-0.29
Forgery	0.04	-0.35	-0.30	-0.17	-0.07	-0.04	0.67	0.34	0.23	0.23
Weapons	-0.17	0.03	-0.10	-0.13	-0.14	0.01	0.13	-0.08	0.38	-0.14
Arson	0.02	0.45	1.06	-2.71	1.86	-0.11	-2.94	2.24	-3.17	0.71
Stolen property	1.46	-4.31^{***}	3.16^{*}	-0.25	1.59^{**}	1.28	-3.74	-3.52	0.95	0.66
Liquor	1.01^{*}	-0.88	1.21	0.36	1.02^{**}	1.03^{**}	-2.82	0.09	-0.40	0.26
Gambling	-0.18	-0.86	-0.60	1.84	-1.72	-0.76	0.37	-1.17	1.05	2.02
Bad checks	-0.27	0.60	-0.91	0.12	0.40	-0.21	0.77	0.57	1.37	0.16
Family offenses	-0.43	0.79	-1.13	-0.73	-0.43	-0.01	1.29	0.25	1.74^{*}	1.17^{*}

Notes. Per-SD interaction gradient (% change in the near–far discount per SD of the tract moderator) for all estimable offenses; continuous interaction, z standardized over the sale sample, fully interacted, each gradient net of the tract's predetermined crime level (also interacted), parcel and tract×month fixed effects, clustered by tract. Columns: Inc=median household income, Pov=poverty rate, Col=% college-plus, Own=% owner-occupied, HV=median home value, Rent=median gross rent, Blk=% Black, His=% Hispanic, Seg=racial segregation (divergence) index, Vac=vacancy rate.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA21: Income and Racial Composition Estimated Jointly

	Model A		Model B	
Offense [window]	Income	Share Black	Income	Segregation
Homicide [10]	0.86	−2.40	2.06*	−1.30
	(1.12)	(2.14)	(1.15)	(1.82)
Burglary [8]	0.21*	−0.45	0.37***	−0.34
	(0.11)	(0.37)	(0.13)	(0.32)
Narcotics [10]	0.21	−0.14	0.41***	0.03
	(0.16)	(0.24)	(0.13)	(0.18)
Vandalism [6]	0.03	−0.44*	0.14	−0.56**
	(0.07)	(0.24)	(0.10)	(0.22)

Notes. Change in the near–far discount (percent) per standard deviation of each tract characteristic, with both characteristics interacted with the near–far term in the same regression, together with the tract's predetermined crime level. Model A: median household income and share Black (non-Hispanic). Model B: median household income and racial segregation (divergence index). Positive values mean a smaller discount. Parcel and tract × month fixed effects; standard errors clustered by tract; $N = 242{,}815$. Window in brackets (months).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

H Inference and Specification Robustness

H.1 Distressed Sales

Foreclosures and short sales could drive the discounts if distressed sellers cluster near crime. Allowing the near–far effect to differ for distressed sales (REO, short sale, or foreclosure), the point estimates are more negative for distressed sales, although these estimates are imprecise and the difference is statistically significant only for narcotics (Table OA22). This comparison describes heterogeneity by distressed status; it does not isolate a forced-seller mechanism (Campbell et al., 2011; Ellen et al., 2013; Anenberg and Kung, 2014). A predetermined vacancy control, described above, leaves the discounts essentially unchanged.

Table OA22: Arm's-Length versus Distressed Sales: Pooled Decomposition

Offense	Window	Arm's-length (%)	Distressed (%)	p(diff)
Homicide	10	-1.78^{*}	-8.99	0.128
		(1.08)	(4.74)	
Burglary	8	-0.12	-0.73	0.254
		(0.10)	(0.52)	
Narcotics	10	-0.00	-0.64	0.026
		(0.13)	(0.26)	
Vandalism	6	-0.09	-0.69	0.186
		(0.07)	(0.45)	
Disorderly conduct	12	-0.95^{***}	0.58	0.469
		(0.36)	(2.09)	
Prostitution	11	-0.29	0.89	0.111
		(0.26)	(0.63)	
Weapons	1	-0.23	-5.01	0.711
		(1.26)	(12.85)	
Arson	4	-2.36	-1.79	0.958
		(2.56)	(10.54)	

Notes. Near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ on ln(sale price), pooled 2007–2025, at each offense's main window (months). Arm's-length and distressed discounts come from one regression with a distress interaction; p(diff) is the Wald test of their equality. Arm's-length = ordinary buyer-chosen sales; distressed = REO, short-sale, or foreclosure (about 13 percent of transactions). Parcel and tract × year-month fixed effects, clustered by tract. Standard errors in parentheses; stars on the arm's-length estimate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

H.2 Reporting and Detection

Two checks bear indirectly on reporting-related confounding. Predetermined 311 service requests reflect both neighborhood conditions and residents' reporting behavior; including them leaves the estimates nearly unchanged (Table A20). Post-sale crime provides a check on persistent local factors, potentially including reporting differences. Across the 63 specifications per offense, post-sale crime is statistically significant at the five-percent level in none for homicide or vandalism, one for narcotics, and four for burglary (Section 6.3). These checks do not directly measure changes in reporting or enforcement. The distinction is particularly relevant for narcotics, whose recorded incidents depend on police detection and enforcement activity (Black, 1970, p. 735).

Spatial standard errors. Conley standard errors use a uniform kernel with cutoffs of 0.5 and 1.0 km. The 1.0 km cutoff exceeds twice the 0.30-mile outer radius used to construct crime exposure, so parcels with overlapping exposure areas can have correlated residuals. Inference and the significance stars in the main tables use the tract-clustered standard errors. Clustering instead by the 77 community areas gives $p = 0.034$ for homicide, 0.043 for burglary, 0.059 for vandalism, and 0.12 for narcotics.

Table OA23: Sale Prices: Alternative Standard Errors

Crime	δ (%)	Tract	Conley 0.5 km	Conley 1.0 km
			standard error	
Homicide	-3.02^{**}	(1.38)	(1.25)	(1.38)
Burglary	-0.24^{**}	(0.12)	(0.11)	(0.12)
Narcotics	-0.29^{**}	(0.14)	(0.13)	(0.17)
Vandalism	-0.17^{**}	(0.08)	(0.07)	(0.08)

Notes. Near–far close-price discount δ (%) for each priced offense, estimated on the full 2007–2025 repeat-sales sample ($N = 242{,}815$) with parcel and tract×month fixed effects. Estimates are the baseline specification of Table 2 (homicide at window 10, burglary 8, narcotics 10, vandalism 6). Column 2 is the baseline standard error clustered by census tract. Columns 3–4 report Conley (1999) spatial standard errors (0.5 km and 1.0 km cutoffs, uniform kernel, great-circle distance), estimated jointly with the fixed effects, with no small-sample adjustment. Significance stars use the tract-clustered standard errors.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA24: Higher-Moment Zero-Exposure Test at the Zero-Crime Mass Point

	Narcotics	Vandalism	Burglary
Panel A. Variance ($\hat{\varepsilon}^2$)			
Raw residual	0.0002	−0.0004	−0.0001
	(0.0007)	(0.0003)	(0.0004)
Winsorized (1/99)	−0.0000	0.0001	−0.0001
	(0.0002)	(0.0002)	(0.0002)
Panel B. Skewness ($\hat{\varepsilon}^3$)			
Raw residual	−0.0004	-0.0015^{**}	-0.0022^{***}
	(0.0012)	(0.0007)	(0.0009)
Winsorized (1/99)	0.0001	−0.0002	-0.0007^{**}
	(0.0004)	(0.0002)	(0.0003)
Observations		242,815	

Notes. Coefficient on the zero-exposure indicator $\mathbf{1}(\text{near} = 0)$ added to the conditioned second and third moments of the price residual (Caetano et al., 2021 higher-moment version), at the baseline PIN + tract×close-month design; standard errors clustered by tract in parentheses. A null coefficient means the moment is continuous at the mass point. The variance, the signature of heterogeneous selection, is a precise null for all three crimes. Raw-residual skewness flags for vandalism and burglary are driven by tail sales; vandalism's is no longer significant after winsorizing, while burglary's remains significant at the five-percent level (−0.0007). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

H.3 Full Specification Curves (All Exposure Windows)

Figures OF4–OF7 report the sale-price specification curve for the four priced offenses across all 103 specifications. The top panel sorts the near–far discount δ (%) across the grid: 72 specifications vary the exposure window (1–12 months), the fixed-effect design (parcel + tract×month, or parcel×era + tract×month), and the far ring (0.20–0.30, 0.25–0.35, 0.30–0.40 mi) under a linear count with no controls; 24 add the full predetermined-control set at the 0.20–0.30 ring; 5 add one control at a time at the baseline specification; and 2 replace the linear count with a presence indicator or

an inverse hyperbolic sine. Navy marks estimates significant and negative at five percent; maroon marks the main specification. The middle panel decodes each specification; the bottom panel is the post-sale placebo. Pooled 2007–2025, clustered by tract.

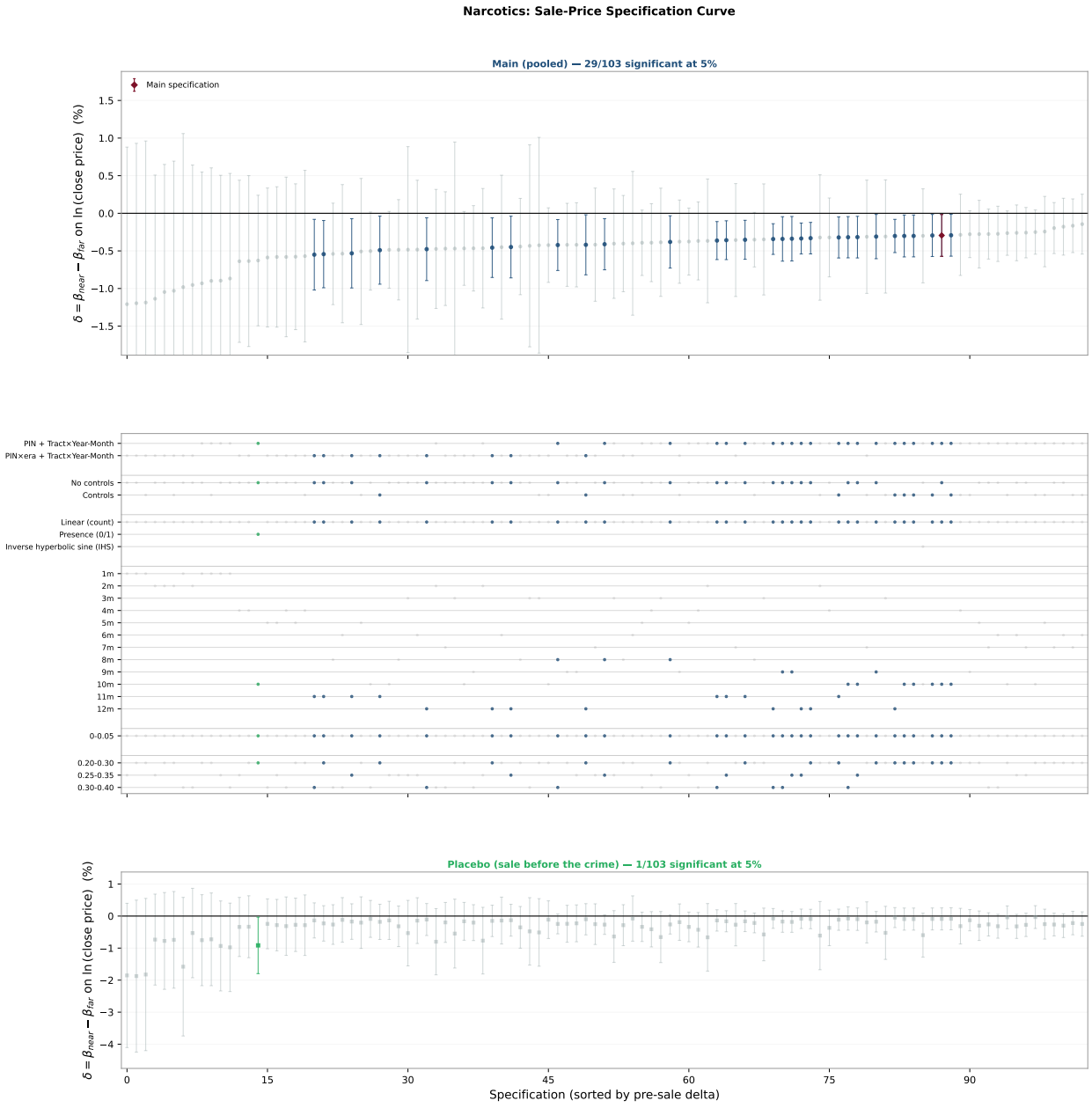


Figure OF4: Narcotics: Sale-Price Discount Across All 103 Specifications. Main: 29/103 significant; placebo 1/103.

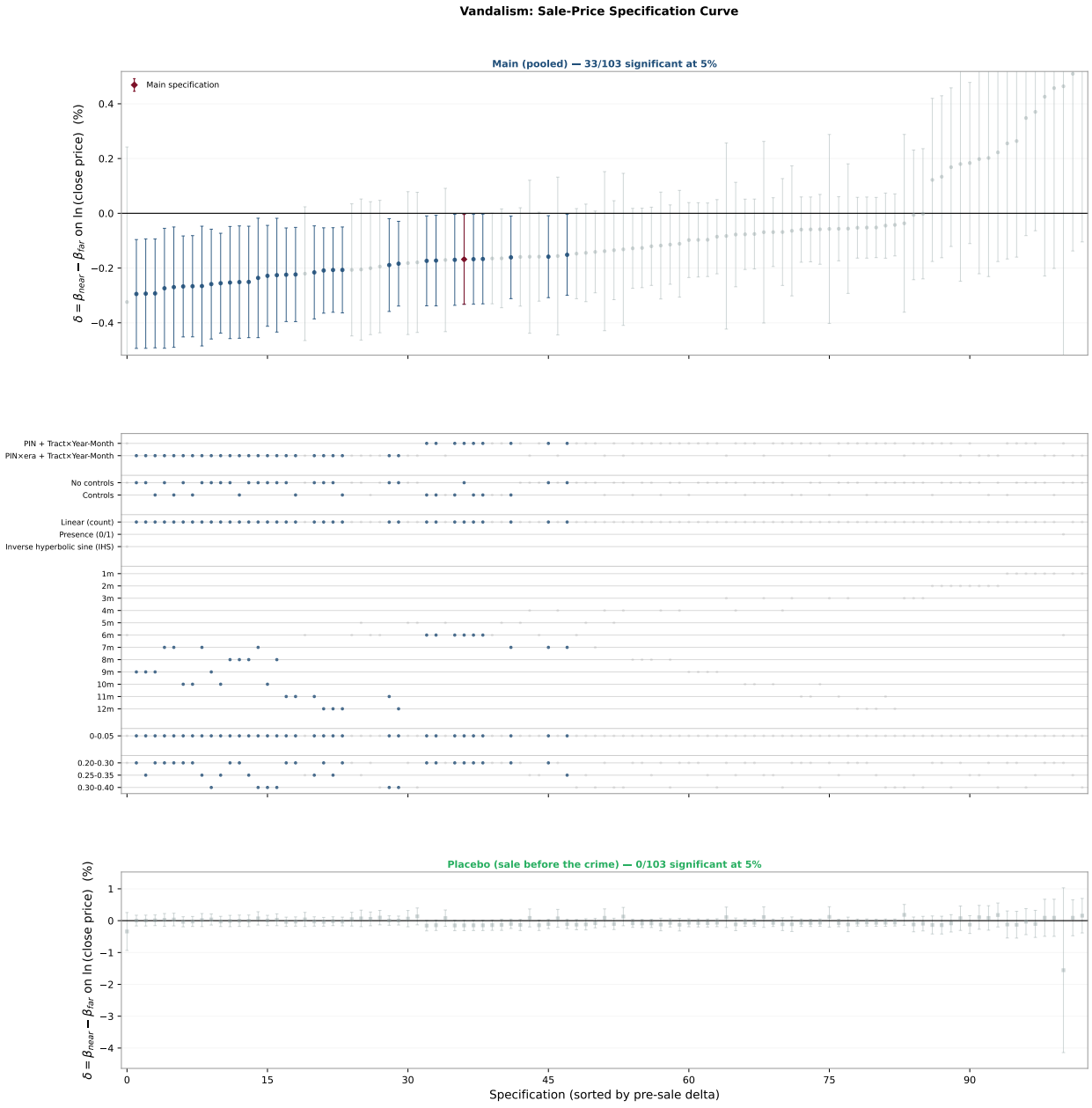


Figure OF5: Vandalism: Sale-Price Discount Across All 103 Specifications. Main: 33/103 significant; placebo 0/103.

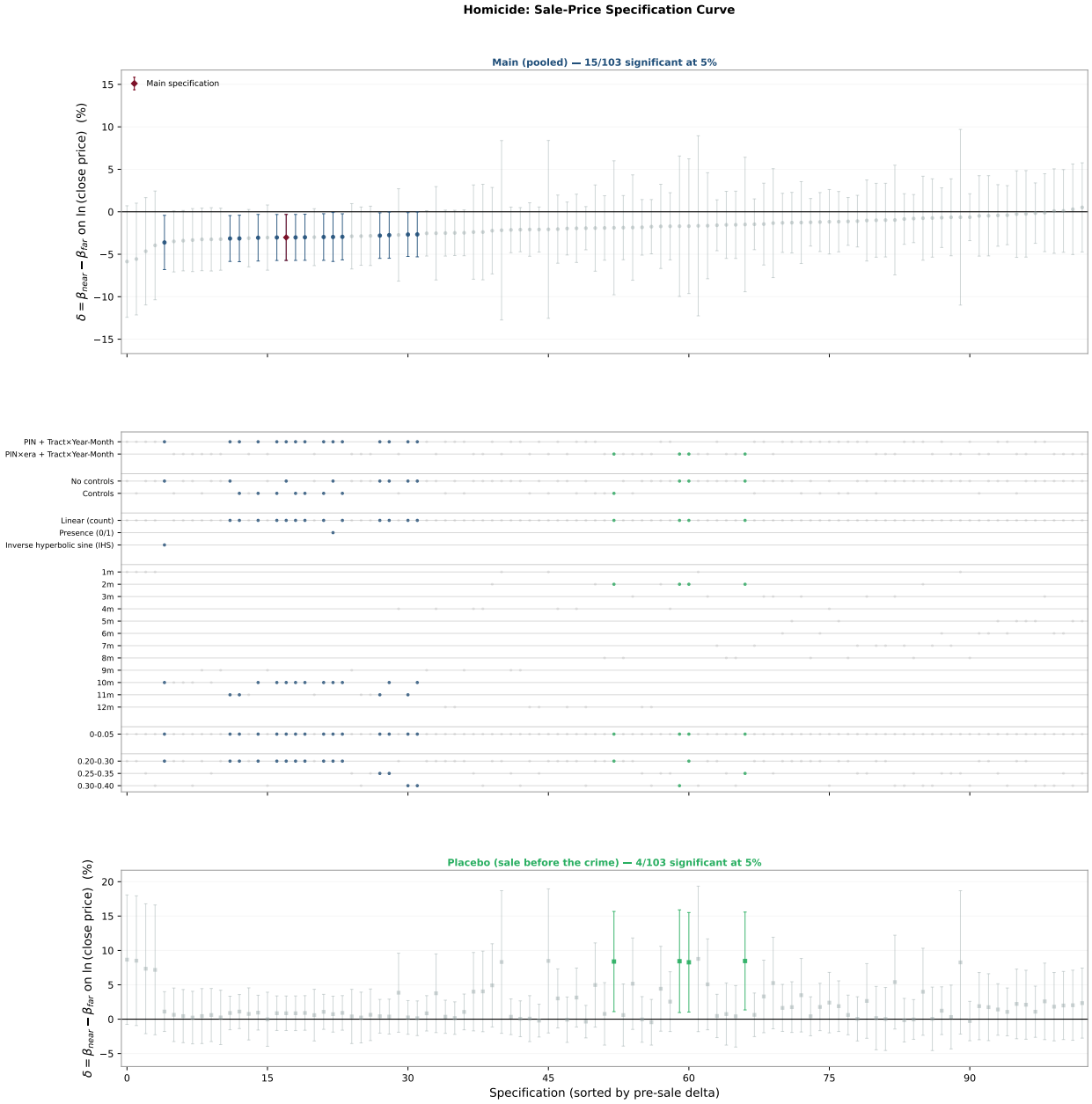


Figure OF6: Homicide: Sale-Price Discount Across All 103 Specifications. Main: 15/103 significant; placebo 4/103, all positive and at the two-month window.

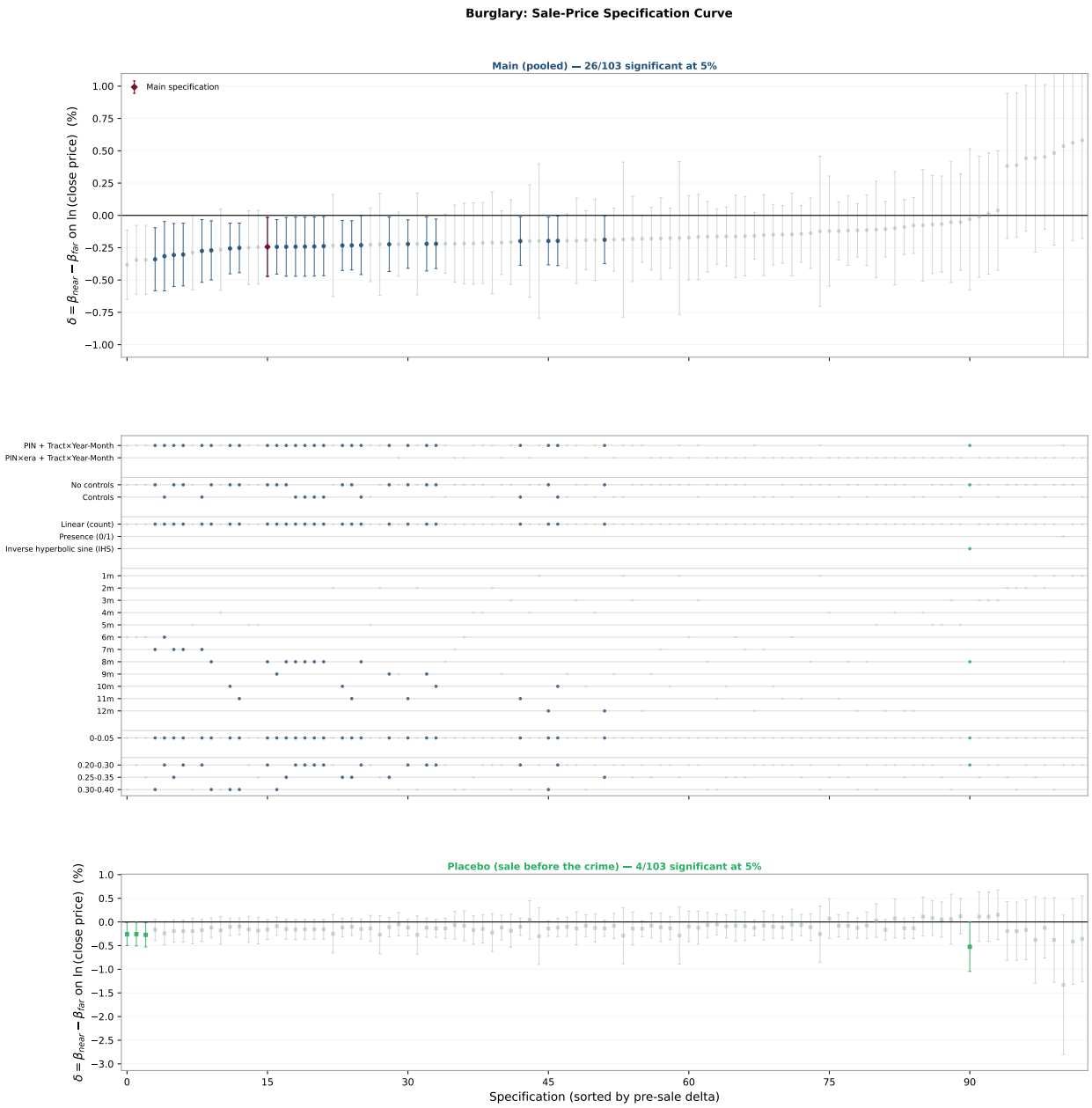


Figure OF7: Burglary: Sale-Price Discount Across All 103 Specifications. Main: 26/103 significant, all in the discount direction; placebo 4/103.

I Functional Form, All Offenses

Table OA25 compares specifications using raw crime counts, a presence indicator, and the inverse hyperbolic sine transformation. A presence indicator treats all positive counts alike, while IHS compresses differences at higher counts. The coefficients consequently have different interpretations: count coefficients summarize the response per additional incident, presence coefficients capture the response to any exposure, and IHS coefficients are expressed per unit of transformed exposure. Where reported as near–far contrasts, each compares the corresponding near- and far-ring coefficients.

At their respective main windows, homicide's discount is significant under all three specifications. Narcotics remains negative throughout (raw count -0.29, presence -0.63, IHS -0.30), but only the raw-count estimate is significant at five percent. Burglary and vandalism are likewise significant only under raw counts. The count-indicator estimates provide a complementary assessment without imposing a constant slope: four or more burglaries and vandalism incidents carry significant discounts (Table OA7). The linear specification provides a per-incident summary, while these estimates show how the response varies across exposure levels.

Across all twenty-two offenses (Table OA25), the choice of functional form does not manufacture the priced discounts; some offenses that do not price in the baseline count specification do turn significant under the inverse-hyperbolic-sine or presence transforms, which rescale or redefine exposure. Three offenses outside the priced set are significant under all three forms: disorderly conduct, negative (-0.65, -1.00, and -0.81, the last at ten percent), and bad checks and family offenses, positive (0.66, 1.06, and 1.07; and 1.00, 1.69, and 1.69, the first at ten percent).

Table OA25: Functional Form, All Estimable Offenses

Offense	Raw count	Presence (≥ 1)	IHS	Window
Homicide	-3.02^{**}	-2.97^{**}	-3.61^{**}	10
	(1.38)	(1.47)	(1.63)	
Burglary	-0.24^{**}	0.54	-0.03	8
	(0.12)	(0.98)	(0.28)	
Narcotics	-0.29^{**}	-0.63	-0.30	10
	(0.14)	(0.44)	(0.32)	
Vandalism	-0.17^{**}	0.46	-0.32	6
	(0.08)	(1.26)	(0.29)	
Theft	0.01	12.01^{*}	-0.15	12
	(0.02)	(6.92)	(0.64)	
Motor-vehicle theft	-0.02	-1.23	-0.35	12
	(0.09)	(0.84)	(0.32)	
Robbery	-0.03	-0.35	-0.25	12
	(0.13)	(0.74)	(0.28)	
Assault (non-dom.)	-0.04	0.28	-0.33	12
	(0.04)	(1.77)	(0.40)	
Assault (dom.)	-0.19^{*}	-0.83	-0.59^{**}	12
	(0.10)	(1.31)	(0.28)	
Disorderly conduct	-0.65^{**}	-1.00^{**}	-0.81^{*}	12
	(0.29)	(0.48)	(0.42)	
Trespass	-0.14	-0.66	-0.52^{*}	12
	(0.10)	(0.59)	(0.28)	
Prostitution	0.08	0.76	0.36	12
	(0.12)	(0.98)	(0.73)	
Fraud	-0.11	-0.16	-0.49	12
	(0.08)	(1.15)	(0.33)	
Sex offenses	-0.47^{**}	-0.66	-0.74^{**}	12
	(0.23)	(0.47)	(0.33)	
Forgery	0.21	0.39	0.26	12
	(0.17)	(0.44)	(0.32)	
Weapons	0.10	0.35	0.19	12
	(0.28)	(0.50)	(0.50)	
Arson	0.07	-0.43	-0.12	12
	(1.03)	(1.08)	(1.19)	
Stolen property	-0.43	-0.72	-0.73	12
	(1.05)	(1.41)	(1.50)	
Liquor	0.36	0.95	0.76	12
	(0.35)	(0.71)	(0.61)	
Gambling	-1.94	-3.39	-3.13	12
	(1.75)	(2.84)	(2.47)	
Bad checks	0.66^{**}	1.06^{**}	1.07^{**}	12
	(0.33)	(0.46)	(0.44)	
Family offenses	1.00^{*}	1.69^{**}	1.69^{**}	12
	(0.51)	(0.71)	(0.69)	

Notes. Near–far close-price effect (%) of each estimable offense's near-ring exposure entering as a raw count, a presence indicator (≥ 1), and inverse hyperbolic sine, each at the offense's main window, pooled 2007–2025, parcel and tract×month fixed effects, clustered by tract. The raw count is the form the dose-response implies (Table OA7). Priced offenses are discussed in the functional-form robustness check (Section 6). Standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

J Nonparametric Distance–Time Surfaces

The near ring and exposure window in the main text are chosen based on a nonparametric estimate of the price effect as a joint function of distance D and time since incident L. Figure OF8 plots the estimated surface $f(D, L)$, a tensor-product spline in distance and lag, on the MLS close-date sample, 2007–2025, for the three offenses with enough incidents to identify a two-dimensional smooth: burglary, narcotics, and vandalism. Homicide is too sparse to support a stable two-dimensional surface. In every case the effect is concentrated at short distances and recent lags and flattens to zero well before the far ring, which is what justifies the near ring and the 0.20–0.30 mile far ring.

Each surface is a tensor-product spline $\text{te}(D, L)$ estimated following Korolev (2026): a signal regression of the price, demeaned by parcel and tract $\times$ month, on the crime count, with the smooth entering through a by term and grid-centered for identification, and the penalty chosen from the data by fast REML. This is a varying-coefficient model: the by term makes the crime count the treatment, and the tensor smooth lets its coefficient vary over distance and lag. The moderators are the geometry of exposure, how far the incident occurred and how recently, rather than subpopulation attributes such as buyer or neighborhood characteristics, so the surface shows how the effect of a given incident varies with the distance and recency of exposure. Following his procedure I add no wiggliness inflation, so the smoothness is what fast REML selects. Because exposure enters as counts on a discrete grid of six distance rings and twelve monthly lags rather than as continuous distances, the tensor bases are correspondingly small, six and four, below the $k = c(10, 10)$ he uses for continuous smooths. The surface is not additively separable: an additive form $f(D) + f(L)$ is rejected against the full interaction by AIC, so distance and time since incident do not enter as a sum. Multiplicative (rank-one) separability is a distinct property, and it holds only descriptively and only relative to the far-ring reference. Referenced to the far ring, matching the reference used in the near–far design, the near-band surface is approximately rank-one for burglary, meaning its spatial decay has the same shape at every lag with only the amplitude fading; for narcotics it is not, the decay shape itself changing with lag. This rank-one share is descriptive, not a formal test. I therefore estimate the full tensor-product surface rather than imposing either restriction.

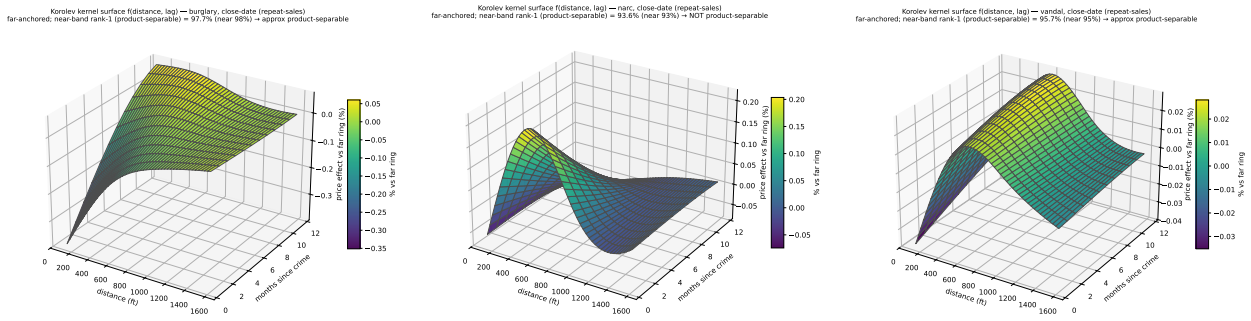


Figure OF8: Estimated price-effect surfaces $f(D, L)$ over distance and time since incident, MLS close-date sample, 2007–2025: burglary (left), narcotics (center), vandalism (right). Each is a tensor-product spline fit over the full distance domain and referenced to the far ring: its value in the 0.20–0.30 mile far ring is set to zero. The effect declines steeply in distance and lag and is close to zero before the far ring. (This far-ring reference is what makes the rank-one comparison in the text well-defined; the main-text distance profile is instead centered over its plotted range, a different reference for a different question.)

The distance profile is fit over 0–0.30 miles, the full domain spanning the far comparison ring, and centered on its average over that range, so it identifies how the effect varies with distance rather than its level, which the fixed effects absorb. The near-ring estimate is essentially invariant across the basis sizes tried, moving by only 0.011 between $c(4, 4)$ and $c(6, 12)$; the reported profile uses the $c(6, 12)$ fit, with one distance basis per 0.05-mile ring across the domain. The profile averages over the one-to-twelve-month exposure window whereas the near–far estimates are reported at each offense's main window, so the profile's homicide level is smaller than the baseline discount rather than in conflict with it; the smooth identifies the level precisely only for offenses with dense exposure, and for rarer offenses the near–far difference carries the precision.

K Cross-Crime Confounding

A natural concern is that a discount attributed to one crime is in fact capitalizing a *different*, spatially correlated crime, that the homicide gradient, say, merely tracks nearby burglary or narcotics activity. Two facts limit this concern for homicide, burglary, and narcotics: the near-ring counts of distinct crimes are mostly weakly correlated, and, entered jointly at a common twelve-month window, each of these offenses retains its own coefficient. Vandalism is the exception.

Near-ring counts are close to orthogonal. Figure OF9 reports the pairwise Pearson correlation of the near-ring (0–0.05 mile) incident counts across all 923,730 MLS listings (all geocoded closed sales before the constant-parcel and repeat-sales filters), for all twenty-two non-sparse offense groups. Homicide is nearly uncorrelated with the other offenses: it correlates at most 0.18 with any other offense group. Among the four priced crimes, the strongest pair, burglary and vandalism, reaches 0.47. Because omitted-variable bias scales with the correlation between the included and the omitted regressor, offenses that are nearly uncorrelated leave little room for one crime's gradient to be attributed to another's: even a large omitted effect, multiplied by a correlation near

0.1, perturbs the estimated coefficient negligibly.

A joint horse-race of all twenty-two offenses. The mechanical argument can be checked directly. Racing only the priced crimes against one another would be selection one step later, so I enter *all twenty-two* offenses' near–far terms in a single regression (common twelve-month window, parcel and tract-by-month fixed effects, clustered by tract, $N = 242{,}815$; Table OA26). For homicide, burglary, narcotics, and disorderly conduct, the discount conditional on the other twenty-one is nearly identical to its separate estimate: narcotics -0.29% ($p = 0.01$) against -0.33% separately, burglary -0.17% against -0.18%, homicide -2.4% against -2.5%, and disorderly conduct -0.58%. The offenses that do not price remain null when entered jointly: larceny/theft, robbery, and motor-vehicle theft stay within hundredths of a percent of zero. Standard errors are essentially unchanged, since the near-ring counts are weakly correlated. Because the horse-race is run at a common twelve-month window, it does not test vandalism's six-month discount: at twelve months vandalism's separate estimate is already small and insignificant (-0.05, s.e. 0.06), and its joint estimate is -0.02.

Where co-movement occurs. The two most correlated pairs in Figure OF9, larceny/theft with fraud (0.70) and larceny/theft with non-domestic assault (0.55), involve only offenses that do not price. Vandalism, however, co-moves with the high-volume offenses: domestic assault (0.55), non-domestic assault (0.51), and other offenses (0.48), and it correlates with burglary at 0.47. Narcotics correlates at 0.41 with domestic assault and 0.43 with gambling. Homicide correlates at most 0.18 with any other offense.

The tighter ring is the conservative choice. Widening the near ring to 0–0.10 mile raises the cross-crime correlations, burglary and vandalism rise from 0.47 to 0.63, because a larger disk pools more of the shared spatial background. The 0–0.05 mile ring used throughout is therefore the specification under which distinct crimes are *most* separable, not the one that most helps the argument. Choosing it makes cross-crime confounding harder to sustain, not easier.

Table OA26: Cross-Crime Horse Race: Each Offense Conditional on the Other Twenty-One

	Separate		Joint (all 22)	
Offense	Discount (%)	(SE)	Discount (%)	(SE)
Homicide	-2.49^{*}	(1.36)	-2.44^{*}	(1.36)
Burglary	-0.18^{*}	(0.09)	-0.17^{*}	(0.09)
Narcotics	-0.33^{***}	(0.11)	-0.29^{**}	(0.11)
Vandalism	-0.05	(0.06)	-0.02	(0.06)
Disorderly conduct	-0.65^{**}	(0.29)	-0.58^{**}	(0.29)
Sex offenses	-0.47^{**}	(0.23)	-0.40^{*}	(0.23)
Assault (domestic)	-0.19^{*}	(0.10)	-0.16	(0.10)
Assault (non-dom.)	-0.04	(0.04)	-0.01	(0.04)
Robbery	-0.03	(0.13)	-0.01	(0.13)
Larceny/theft	0.01	(0.02)	0.02	(0.02)
Motor vehicle theft	-0.02	(0.09)	-0.00	(0.08)
Trespass	-0.14	(0.10)	-0.07	(0.09)
Fraud	-0.11	(0.08)	-0.14^{*}	(0.08)
Weapons	0.10	(0.28)	0.20	(0.28)
Arson	0.07	(1.03)	0.15	(1.03)
Prostitution	0.08	(0.12)	0.11	(0.11)
Forgery	0.21	(0.17)	0.23	(0.17)
Bad checks	0.66	(0.33)	0.72	(0.34)
Liquor	0.36	(0.35)	0.56	(0.37)
Family	1.00	(0.51)	1.12	(0.51)
Gambling	-1.94	(1.75)	-1.10	(1.85)
Stolen property	-0.43	(1.05)	-0.48	(1.03)

Notes. Each offense's near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ (percent sale-price change per near-ring incident), at a common twelve-month exposure window, pooled 2007–2025, with parcel and tract × year-month fixed effects, clustered by tract (N = 242,815, 679 clusters). The *Separate* columns estimate one offense at a time; the *Joint* columns enter all twenty-two offenses' near and far terms in a single regression, so each discount is conditional on the other twenty-one. Reported as a robustness exhibit; standard errors are essentially unchanged in the joint specification. Stars from the joint p-value, shown only for discounts.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

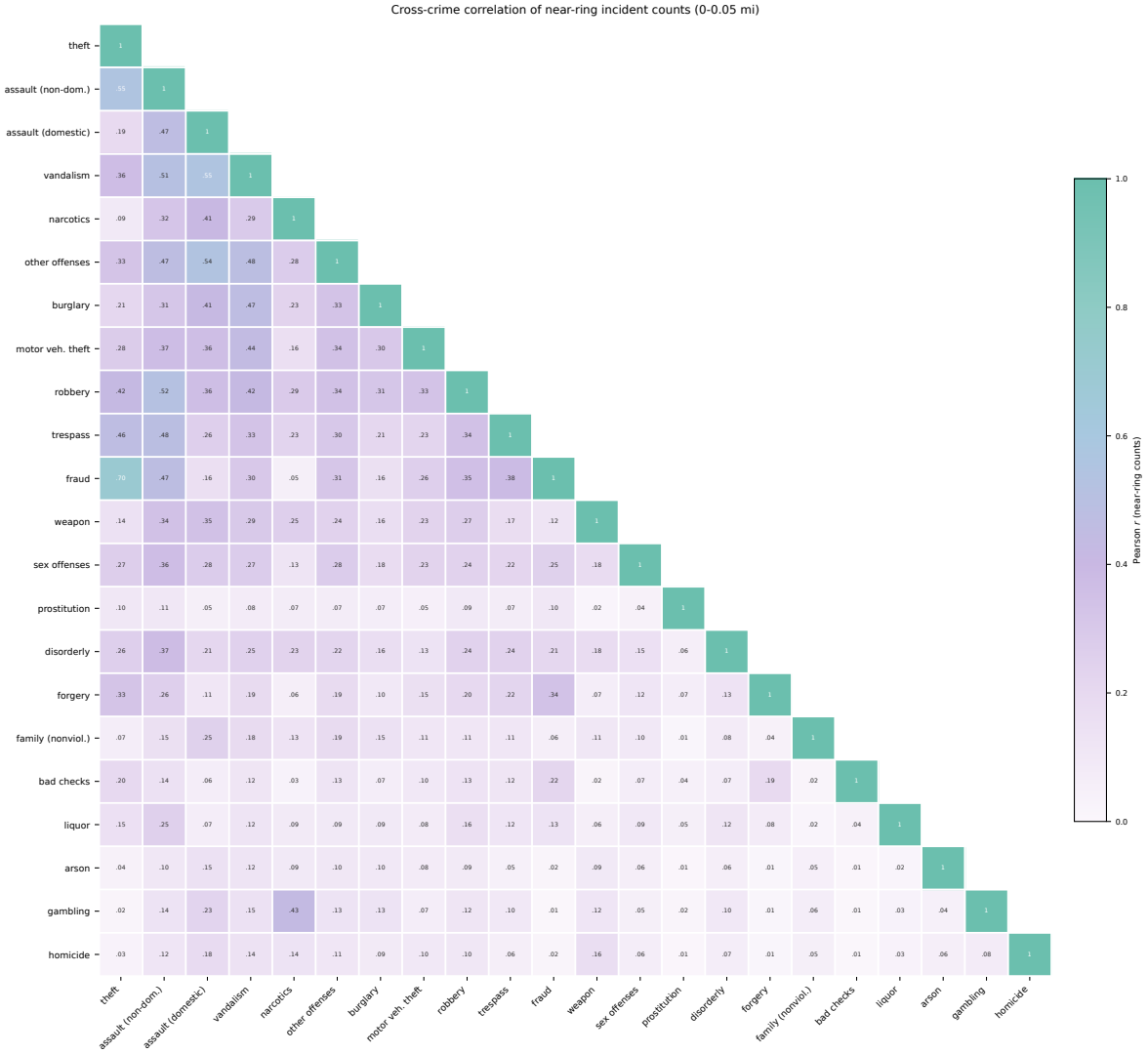


Figure OF9: Cross-crime correlation of near-ring incident counts. Pairwise Pearson correlation of the 0–0.05 mile incident counts over the twelve months before the close of the twenty-two non-sparse offense groups, across 923,730 MLS listings. Homicide is nearly uncorrelated with the other offenses (correlations of at most 0.18); burglary and narcotics are moderately correlated with them (at most 0.47 and 0.43); vandalism is more strongly correlated, especially with domestic and non-domestic assault (0.55 and 0.51). The only strongly correlated pairs (larceny/theft–fraud, larceny/theft–assault) are among the crimes that do not price. The wider 0–0.10 mile ring raises these correlations, confirming that the 0–0.05 mile ring is the more separable specification.

L Does Repeat-Sales Selection Drive the Discount? A Heckman Correction

The capitalization estimates are identified from repeat sales: a parcel enters the identifying sample only if it transacts at least twice, so that its fixed effect can be differenced out. This design invites an obvious objection. If nearby crime changes *which* parcels resell, the sample of twice-sold homes is endogenously selected, and the estimated discount could reflect who-sells rather than what-crime-does-to-price. I confront this directly with a Gatzlaff–Haurin two-equation selection correction (Gatzlaff and Haurin, 1997; Heckman, 1979), adapted to the MLS close-date panel.

A selection model with an excludable instrument. Selection operates through two transaction margins, each estimated as an annual-risk-set probit. A *first-close* probit models whether a parcel records its first MLS sale in a given year (covariates: living area, lot size, age, unemployment); a *second-close* probit models whether a previously-sold parcel resells (adding the excluded instrument). From each probit I form the inverse-Mills ratio $m = \phi(Z\pi)/\Phi(Z\pi)$ and attach it to the corresponding transaction. Because a home's first and second sales draw their m from *different* probits, the selection term varies within a parcel and is not swept away by the parcel fixed effect, a subtle but essential feature that makes the correction identified in a repeat-sales design.

The excluded instrument is mortgage-rate *lock-in*: the gap between the prevailing thirty-year rate at a potential resale and the rate the owner locked in at purchase, $r_{30}(t) - r_{30}(t_{\text{prior}})$. Lock-in strongly governs *whether* an owner sells (surrendering a below-market mortgage is costly, so a large positive gap suppresses mobility), but it bears no plausible relation to the *cross-sectional near-versus-far crime gradient* realized at the moment of sale. It shifts selection without touching the price equation.

Results. Table OA27 delivers the sharp result. The inverse-Mills coefficient is large, positive, and significant at the 0.1% level for every priced crime: the repeat-sales sample *is* selected, and the correction has real bite. This matters because it means the test is not vacuous: the data do exhibit the selection the objection worries about, and the Heckman term is capable of moving the estimate. Adding m changes the point estimates little: homicide shifts from -4.35% to -4.18%, narcotics from -0.47% to -0.52%, vandalism from -0.40% to -0.34%, and burglary from -0.36% to -0.33%; burglary is not significant either before or after the correction in this smaller sample.

The logic is what makes this convincing. A null correction is uninformative when the instrument is weak or selection is absent; one learns nothing from a test with no power. Here selection is strongly present (m highly significant) and is explicitly purged, and the point estimates change little. The composition of the twice-sold sample is not random, m says as much, but that selection is *orthogonal to the near-versus-far crime gradient*: it shifts the price level, not the crime slope. For homicide, narcotics, and vandalism, the discounts are therefore not explained by which homes resell; for burglary the check is uninformative because the estimate is imprecise in this sample.

Table OA27: Capitalization Under a Repeat-Sales Selection Correction

Crime	Uncorrected δ (%)	Selection-corrected δ (%)	Selection term m (Mills)
Homicide	-4.35^{**}	-4.18^{**}	0.72^{***}
Burglary	-0.36	-0.33	0.72^{***}
Narcotics	-0.47^{***}	-0.52^{***}	0.72^{***}
Vandalism	-0.40^{**}	-0.34^{**}	0.72^{***}

Near = 0–0.05 mi, far = 0.20–0.30 mi; outcome ln(close price); PIN and tract×close-month fixed effects; standard errors clustered by tract; pooled sample, each offense at a fixed window (homicide 10, burglary 8, narcotics 12, vandalism 7 months), which differ from the main windows for narcotics and vandalism, with a persistence control for near-ring crime beyond the window cumulated to twelve months. m is the inverse-Mills ratio from a two-probit Gatzlaff–Haurin selection model (first-close and second-close probits; excluded instrument = mortgage-rate lock-in, plus unemployment). The coefficient on m is nearly identical across offenses (0.717 to 0.720) because the selection equations are driven by house characteristics, lock-in, and unemployment rather than by crime. "Uncorrected" and "selection-corrected" run on the identical m-matched repeat-close sample (N = 130,564) for comparability, smaller and differently selected than the full repeat-sales sample of the main results, so the uncorrected homicide estimate here (−4.35) exceeds the baseline repeat-sales discount (−3.0 percent); the object of the test is the movement from uncorrected to corrected, not the level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

M Is Nearby Crime Exogenous? A Zero-Exposure Test

Burglary details. For burglary the zero-exposure indicator is itself significant (−0.978, s.e. 0.317), so the linear discount is identified partly from the contrast between exposed and unexposed parcels. The discontinuity appears only once parcel fixed effects are included, so it does not reflect burglars targeting more valuable houses, a concern Gibbons (2004) raises for London. The jump does not reflect spillover from the adjacent ring: controlling for burglary at 0.05–0.10 miles leaves it unchanged (−0.99, s.e. 0.32). Because the test jointly detects failures of exogeneity, functional form, and parallel trends, its rejection does not establish the direction of any bias, and for burglary the baseline linear specification is an approximation, not a bound; adding the zero-exposure indicator moves δ from −0.244 (s.e. 0.117) to −0.552 (s.e. 0.168) (Figure OF11). At a 0–0.10 mile near ring (Figure A2), where 27 percent of sales have no nearby burglary and the zero-exposure test does not reject (−0.06, s.e. 0.27), burglary's discount remains significant (−0.14, s.e. 0.07; Table A6), with no significant post-sale placebo at any window.

The capitalization estimates identify off within-parcel variation: a home enters the identifying sample only through repeated sales, and the near–far discount is the price movement associated with a change in nearby crime, net of the parcel and neighborhood–month fixed effects. Appendix L shows the estimate is robust to selection into the twice-sold sample. Here I test a distinct assumption, that the near-ring crime count is itself *exogenous* conditional on the design, i.e. uncorrelated with unobserved determinants of price. I do so with a zero-exposure test that requires no exclusion restriction, exploiting a feature of the data: the near-ring count has a *mass point* at zero, because

most homes record no incident in the exposure window.

The zero-exposure test. I implement the test of Caetano et al. (2021). I augment the baseline near–far specification (Equation (5)) with a zero-exposure indicator, equal to one when the near ring records no incident, under the same fixed effects:

$$\ln(\text{close})_{it} = \beta_{\text{near}}\text{Near}_{it} + \beta_{\text{far}}\text{Far}_{it} + (\text{persistence}) + \gamma\,\mathbf{1}(\text{Near}_{it} = 0) + \mu_i + \lambda_{c(i)t} + \varepsilon_{it}, \quad (1)$$

where μ_i is the parcel fixed effect and $\lambda_{c(i)t}$ the tract×month fixed effect of Equation (5), and I test whether $\gamma = 0$. The logic, due to Caetano (2015) and applied to crime counts with a mass at zero by Caetano and Maheshri (2018) to select among dynamic-spillover models of Dallas police data, is that confounders tend to be *discontinuous* at a mass point: if the zero-crime homes lie on the smooth dose–response traced by the crime-exposed homes, the indicator is zero; a nonzero coefficient signals that the mass-point observations are selected. This regression version dominates the original nonparametric test: it is more powerful, applies to discrete or mixed treatments, and, in a two-way fixed-effects design, jointly detects failures of exogeneity, functional form, *and* the "strong" parallel-trends condition (parallel trends plus uncorrelated treatment effects, the assumption whose failure drives the negative-weights problem).

Which crimes can be tested. The test needs positive support beyond the mass point, homes at one, two, three or more nearby incidents, to trace the dose–response. Homicide fails this: its near ring is ≈ 99% zero, with essentially no observations above one (Figure OF10); arson and stolen property are similarly sparse (not shown), so the test is not identified for them; they are, by construction, a presence contrast rather than a dose. Narcotics, vandalism, and burglary have workable mass points (48–75% zero, with 12–27% of homes at two or more) and are tested.

Selection into zero exposure. Table OA28 reports the coefficient on the zero-exposure indicator as fixed effects escalate. Uncontrolled, it is large and highly significant: for narcotics, zero-crime homes sell for about 50% (a 40.5 log-point coefficient on the zero-exposure indicator) more than the dose–response predicts, and they differ so much on unobservables that the raw near–far slope even flips positive. Tract×close-month fixed effects shrink it by an order of magnitude, and under the baseline PIN + tract×close-month design it is a precise null for both narcotics and vandalism (Figure OF11). Crucially, the discount δ is essentially unchanged whether or not the zero-exposure indicator is included (Panel B). The within-property, within-neighborhood-month design thus absorbs the cross-sectional discontinuity that the raw data plainly exhibit; the diagnostics provide supporting evidence that the estimated discount is not driven by *which* homes have no nearby crime.

Burglary. The zero-exposure and skewness tests both indicate that zero-exposure parcels differ for burglary. Two estimates identify the slope from exposed parcels only: adding the zero-exposure

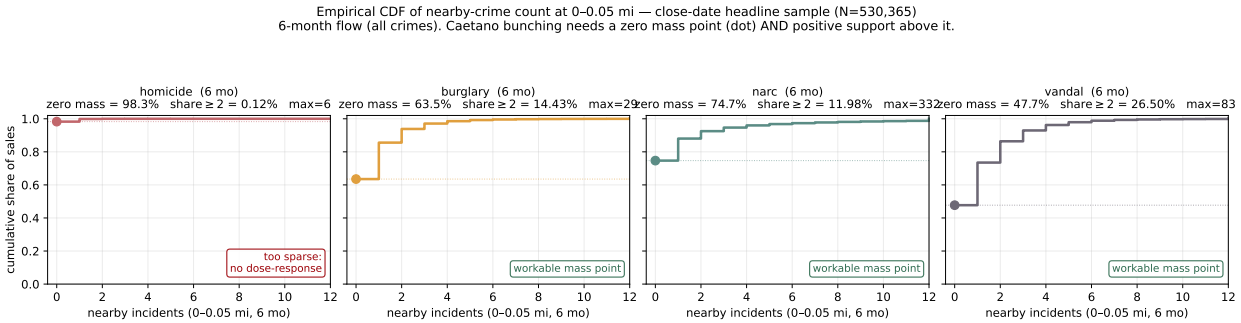


Figure OF10: Which crimes can be tested: empirical CDF of the near-ring count. The zero-exposure test needs a zero mass point *and* positive support above it. Homicide is 99% zero, too few properties with any nearby incident; narcotics, vandalism, and burglary have workable mass points.

indicator gives −0.552 (s.e. 0.168), and the binned dose-response (Table OA7) gives 0.48 (s.e. 0.25) for a single incident and −1.85 (s.e. 0.76) for four or more, a cumulative rather than per-incident effect. The rejection does not establish the direction of any remaining bias, so the baseline linear specification, −0.244 (s.e. 0.117), is an approximation, not a bound. The jump is not spillover from the adjacent ring (controlling for 0.05–0.10 burglary leaves it at −0.99, s.e. 0.32), and at a 0–0.10 near ring the test does not reject (−0.06, s.e. 0.27) while the discount remains significant (−0.14, s.e. 0.07).

Higher moments. Finally, I apply the same zero-exposure test to the second and third moments of the price residual, conditioned on the covariates (Figure OF12 and Table OA24). A discontinuity in the *variance* at the mass point would signal *heterogeneous* selection, the correlated random-effects failure that also invalidates two-way fixed effects. Adding $\mathbf{1}(\text{near} = 0)$ to the conditioned second moment yields a precise null for all three crimes (burglary $p = 0.64$, narcotics 0.99, vandalism 0.51; winsorized, at the baseline parcel and tract×close-month design, 0–0.05 mile, counts at each offense's main window): the diagnostics do not detect a variance discontinuity at zero exposure. Winsorization removes the skewness rejection for vandalism, but burglary's remains significant at the five-percent level ($p = 0.02$). For burglary, the skewness rejection is consistent with the selection at zero exposure that the zero-exposure indicator detects. Identifying the slope from exposed parcels only gives −0.552 (s.e. 0.168); because the test does not establish the direction of bias, the baseline linear specification (−0.244) is an approximation, not a bound.

Table OA28: Zero-Exposure Test of Identification (Caetano–Caetano–Fe–Nielsen 2021)

	Narcotics	Vandalism	Burglary
Panel A. Zero-exposure indicator $\mathbf{1}(near = 0)$*, coefficient (%)*			
Raw (no fixed effects)	40.500^{***}	4.924^{***}	-1.353
	(2.830)	(1.212)	(1.176)
Tract × close-month	4.314^{***}	2.518^{***}	0.329
	(0.784)	(0.578)	(0.538)
PIN + tract × close-month (baseline)	-0.129	-0.323	-0.978^{***}
	(0.386)	(0.256)	(0.317)
+ quadratic dose-response	-0.061	-0.580^{*}	-1.057^{**}
	(0.493)	(0.297)	(0.442)
Panel B. Near–far discount δ *(%), baseline design*			
δ, without dummy	-0.294^{**}	-0.168^{**}	-0.244^{**}
	(0.142)	(0.084)	(0.117)
δ, with dummy	-0.308^{*}	-0.246^{**}	-0.552^{***}
	(0.157)	(0.107)	(0.168)
Observations		242,815	

Notes. The zero-exposure test of Caetano et al. (2021) adds a zero-exposure indicator, $\mathbf{1}(\text{near} = 0)$, to $\ln(\text{close}) = \beta_1 \text{near} + \beta_2 \text{far} + \text{persistence} + \varepsilon$; under valid identification its coefficient is zero. In two-way fixed-effects designs the test jointly detects failures of exogeneity, linearity, and strong parallel trends. Near = 0–0.05 mi, far = 0.20–0.30 mi, each offense at its main window (narcotics 10, vandalism 6, burglary 8 months), with a persistence control for near-ring crime in the remaining months up to twelve; single-parcel sample; standard errors clustered by tract in parentheses. The large raw discontinuity (narcotics 40.5 log points, about 50%: zero-crime homes are dramatically different uncontrolled) is progressively absorbed by fixed effects and is a precise null under the baseline PIN + tract×close-month design; the discount δ is stable with and without the zero-exposure indicator for narcotics and vandalism. The zero-exposure and skewness tests both indicate that zero-exposure parcels differ for burglary. Two estimates identify the slope from exposed parcels only: adding the zero-exposure indicator gives −0.552 (s.e. 0.168), and the binned dose-response (Table OA7) gives 0.48 (s.e. 0.25) for a single incident and −1.85 (s.e. 0.76) for four or more, a cumulative rather than per-incident effect. The rejection does not establish the direction of any remaining bias, so the baseline linear specification, −0.244 (s.e. 0.117), is an approximation, not a bound. The jump is not spillover from the adjacent ring (controlling for 0.05–0.10 burglary leaves it at −0.99, s.e. 0.32), and at a 0–0.10 near ring the test does not reject (−0.06, s.e. 0.27) while the discount remains significant (−0.14, s.e. 0.07). Homicide, arson, and stolen property are not testable (near-ring ≈ 99% zero: too few properties with any nearby incident). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

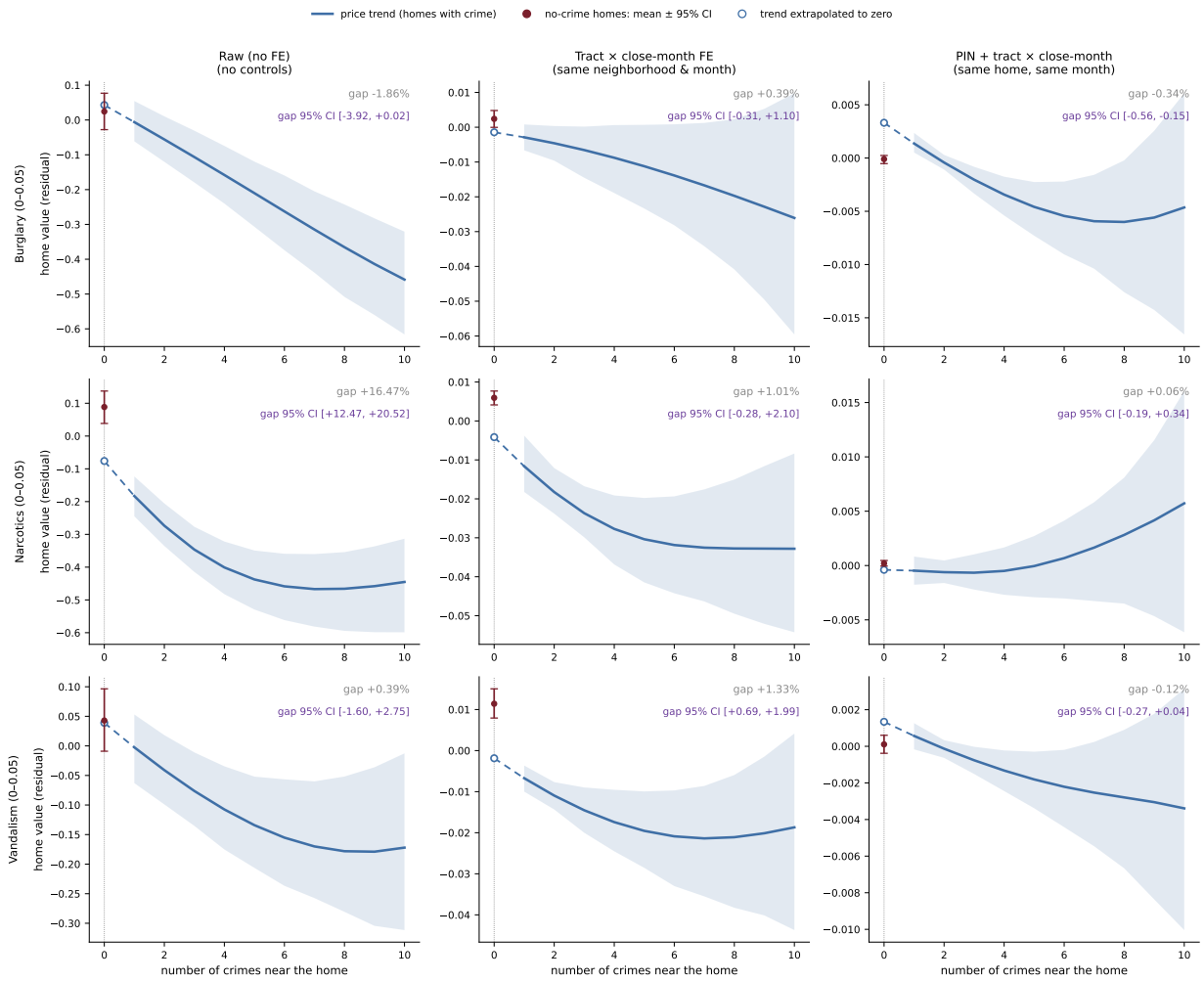


Figure OF11: The zero-exposure diagnostic. Blue: dose–response among crime-exposed homes; red: no-crime homes; open circle: the trend extrapolated to zero. For narcotics and vandalism the large gap in the raw column collapses to zero under the baseline PIN + tract×close-month design; for burglary a gap remains, so its discount is identified partly from the exposed-versus-unexposed contrast; the rejection does not establish the direction of any bias (Table OA28). With a 0–0.10 mile near ring, the burglary gap disappears (−0.06, s.e. 0.27) and the discount remains (−0.14, s.e. 0.07; Section 5.1).

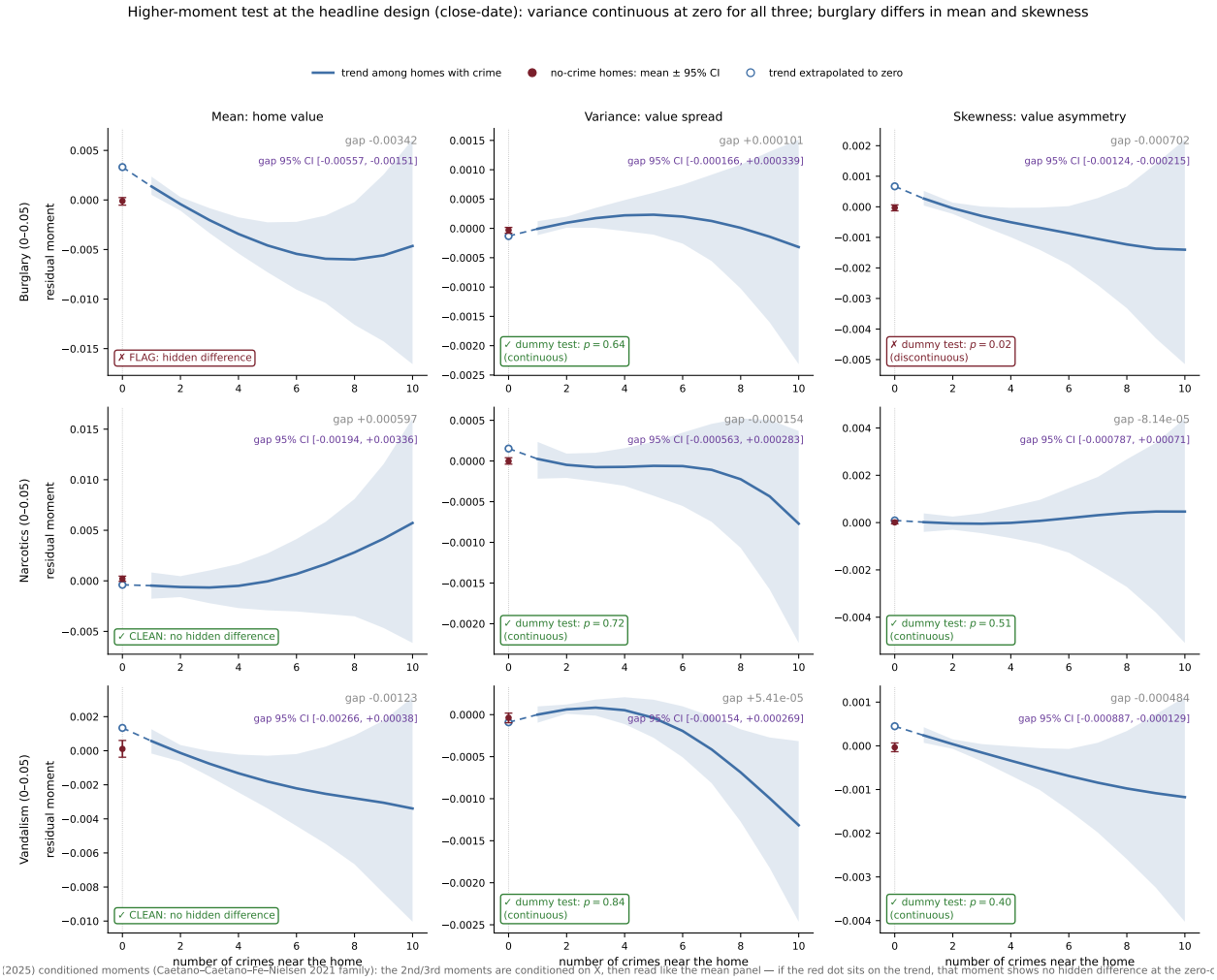


Figure OF12: Higher-moment test. The mean, variance, and skewness of the conditioned price residual at the zero-crime mass point. The parametric zero-exposure test finds the *variance*, the signature of heterogeneous selection, continuous at zero for all three crimes (burglary $p = 0.64$, narcotics 0.99, vandalism 0.51); under winsorizing the skewness rejection is removed for vandalism but not for burglary. The local-linear gaps shown are boundary-sensitive and are superseded by the zero-exposure test.

N Market Tightness, Contract Timing, and Distressed Sales

A nearby incident can lower the recorded sale price by revising what buyers believe the home is worth, or by giving a buyer under contract leverage to reopen the agreed price. Both move the close price in the same direction, and a transaction reveals only the equilibrium price, so the baseline regression cannot separate them. I look for the signature of bargaining on the margin where a leverage account differs most sharply from revaluation, the buyer's market power.

I assign each sale the months of supply (active inventory over trailing three-month sales) in its community area in the month it was listed. Because it is measured in the month of listing, it is predetermined relative to the close. It is a standard measure of market tightness, defined within year so that a soft market is soft relative to its own year rather than to the 2009–2012 downturn. A buyer in a soft market holds outside options and leverage that a market of competing offers removes, so a bargaining channel predicts a larger discount where supply is high. Estimating the discount across terciles of supply, with the parcel and tract-by-month fixed effects and stale-crime controls of the baseline specification, I find no such gradient (Table OA29). For burglary, narcotics, and vandalism the market-tightness comparisons do not detect the larger discount in soft markets that bargaining predicts: at no exposure window is the discount significantly larger in soft markets; the only significant differences, for burglary at one and two months and vandalism at one month, show larger discounts in tight markets. Homicide is imprecise: at the twelve-month window its estimate is -4.8 percent in soft markets against -0.3 in tight, but the difference is not significant ($p = 0.13$) and the ordering reverses at shorter windows, so the homicide comparison is inconclusive. A tercile split defined over the whole sample period does produce an apparent gradient, but it places most soft-market sales in the 2009–2012 crisis; the gradient is an era effect that the within-year definition removes.

Two more direct tests are available but lack power in these data. The first compares crime in the twelve months before the contract date, which is embedded in the agreed price, with crime between contract and close, which arrives after the price is set and can move it only through renegotiation. Pre-contract exposure carries the usual discounts, about 0.2 percent per burglary and 0.3 percent per narcotics incident. Post-contract exposure is insignificant for every offense, but its standard errors are too wide to be informative: the burglary and narcotics estimates carry standard errors of 0.24 and 0.58, and the homicide estimate 3.3, so each interval spans effects as large as the belief-channel discount. The window between contract and close has a median of 42 days (mean 48), so few incidents fall in it, and holding that window to a fixed length requires restricting to slow closings, which selects troubled deals and no longer reproduces the belief channel itself. The second test compares distressed sales with arm's-length sales, but distressed transactions are about thirteen percent of the sold sample (REO, short-sale, or foreclosure, the same definition as Table OA22) and their estimates are too dispersed to sign.

The market-tightness comparisons do not show the larger soft-market discount that bargaining predicts. The short-horizon renegotiation and distressed-sale margins remain beyond the reach of these data.

Table OA29: Crime Discounts by Market Tightness

Panel A

	Homicide				Burglary			
Window	Tight	Balanced	Soft	p	Tight	Balanced	Soft	p
1	−12.24**	−2.72	−4.61	0.42	−1.16***	0.51	0.47	0.04
	(6.03)	(6.78)	(6.75)		(0.43)	(0.50)	(0.65)	
2	−6.09	−2.28	−0.07	0.37	−0.95***	0.08	0.30	0.02
	(4.32)	(4.23)	(4.99)		(0.29)	(0.35)	(0.44)	
3	−4.63	−1.19	0.88	0.34	−0.68***	0.20	−0.04	0.15
	(3.72)	(3.99)	(4.50)		(0.23)	(0.29)	(0.38)	
4	−4.24	−1.45	0.18	0.34	−0.59***	0.09	−0.08	0.20
	(3.13)	(3.63)	(3.63)		(0.21)	(0.25)	(0.35)	
5	−5.19**	−1.41	1.60	0.12	−0.46**	−0.03	−0.30	0.67
	(2.64)	(3.11)	(3.41)		(0.19)	(0.23)	(0.31)	
6	−3.21	−2.21	1.16	0.29	−0.48***	−0.15	−0.46*	0.96
	(2.72)	(2.91)	(3.06)		(0.18)	(0.22)	(0.27)	
7	−2.89	−1.92	0.12	0.41	−0.33**	−0.15	−0.51**	0.51
	(2.29)	(2.76)	(2.64)		(0.16)	(0.21)	(0.24)	
8	−2.15	−1.32	−0.29	0.59	−0.31**	−0.11	−0.35	0.87
	(2.28)	(2.66)	(2.42)		(0.15)	(0.18)	(0.23)	
9	−1.78	−1.96	−2.88	0.73	−0.29**	−0.14	−0.21	0.74
	(2.11)	(2.50)	(2.28)		(0.14)	(0.16)	(0.22)	
10	−1.39	−2.58	−3.83*	0.42	−0.27**	−0.15	−0.24	0.86
	(2.02)	(2.47)	(2.19)		(0.13)	(0.15)	(0.20)	
11	−0.90	−1.83	−4.72**	0.21	−0.26**	−0.16	−0.26	0.99
	(1.86)	(2.36)	(2.35)		(0.13)	(0.14)	(0.19)	
12	−0.33	−0.49	−4.77**	0.13	−0.20*	−0.14	−0.21	0.96
	(1.99)	(2.32)	(2.17)		(0.12)	(0.13)	(0.18)	

Panel B

	Narcotics				Vandalism			
Window	Tight	Balanced	Soft	p	Tight	Balanced	Soft	p
1	−0.46	−1.07	−1.30	0.58	−0.22	0.29	0.88*	0.05
	(1.34)	(1.11)	(0.93)		(0.32)	(0.32)	(0.47)	
2	0.15	−0.31	−0.78	0.33	−0.02	0.11	0.36	0.32
	(0.74)	(0.62)	(0.64)		(0.23)	(0.22)	(0.31)	
3	0.06	−0.31	−0.34	0.61	−0.06	0.04	−0.18	0.67
	(0.60)	(0.51)	(0.60)		(0.17)	(0.18)	(0.23)	
4	−0.22	−0.41	−0.36	0.84	−0.12	0.01	−0.23	0.65
	(0.51)	(0.38)	(0.45)		(0.14)	(0.15)	(0.19)	
5	−0.37	−0.43	−0.27	0.87	−0.18	−0.07	−0.24	0.76
	(0.45)	(0.37)	(0.35)		(0.14)	(0.14)	(0.17)	
6	−0.38	−0.25	−0.11	0.57	−0.19	−0.07	−0.27*	0.68
	(0.40)	(0.29)	(0.26)		(0.13)	(0.12)	(0.15)	
7	−0.29	−0.27	−0.13	0.68	−0.15	−0.03	−0.30**	0.40
	(0.35)	(0.26)	(0.21)		(0.11)	(0.11)	(0.14)	
8	−0.39	−0.44*	−0.28	0.75	−0.15	−0.00	−0.22	0.70
	(0.32)	(0.26)	(0.20)		(0.11)	(0.10)	(0.14)	
9	−0.24	−0.40*	−0.23	0.99	−0.09	−0.02	−0.17	0.56
	(0.29)	(0.22)	(0.19)		(0.10)	(0.10)	(0.13)	
10	−0.22	−0.39*	−0.26	0.90	−0.06	−0.03	−0.14	0.56
	(0.27)	(0.22)	(0.17)		(0.10)	(0.09)	(0.12)	
11	−0.25	−0.41**	−0.34**	0.75	−0.07	−0.03	−0.08	0.90
	(0.25)	(0.20)	(0.16)		(0.09)	(0.08)	(0.11)	
12	−0.27	−0.38**	−0.32**	0.84	−0.06	−0.03	−0.08	0.88
	(0.23)	(0.16)	(0.14)		(0.08)	(0.08)	(0.11)	

Notes. Near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$, percent per incident, estimated separately by tercile of months of supply, an inventory-based proxy for the buyer–seller ratio that defines market tightness in search models (Genesove and Han, 2012). Months of supply in a community area and month is the number of active listings (on the market at any point in the month) divided by average monthly closed sales over that month and the two preceding months; each sale is assigned the value for its community area in its original listing month. Terciles are defined within each listing year, so they compare markets within the same year rather than at fixed thresholds: tight (lowest third, a seller's market), balanced, and soft (highest third, a buyer's market). One regression per offense and window with tercile-specific near and far terms, tercile main effects, stale-crime controls, parcel and tract×month fixed effects; standard errors clustered by tract in parentheses. p: Wald test that the soft and tight discounts are equal. Main exposure windows are 10 months for homicide and narcotics, 8 months for burglary, and 6 months for vandalism. N = 232,873 sales with a months-of-supply value (the baseline sample is 242,815). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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